

MEDIATEK

MT6363 Design Notice

for MT8676

V0.1

Document Revision History

Revision	Date	Description
V0.1	2024/02/02	Initial.

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MT6363 Design Notice

MT6363 Introduction

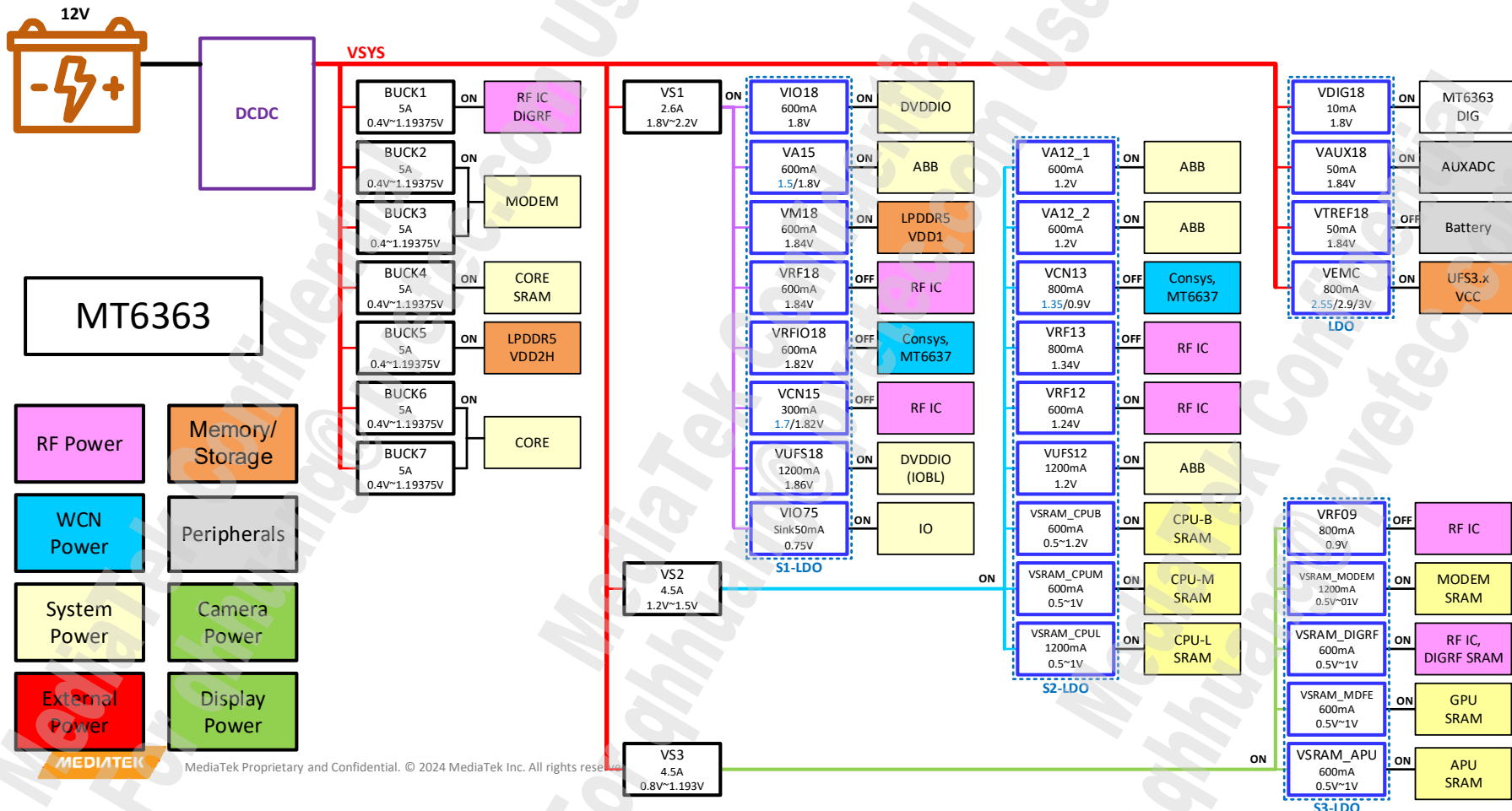
MT6363 – General Description

- The MT6363 is highly integrated with functions fulfilling all power requirements in smart phone system.
 - **Buck converters*10**
 - VS1, VS2, VS3, BUCK1 ~ 7
 - **LDOs*26**
 - **“GPIO + AUXADC”*7**
 - GPIO function or AUXADC function can be chosen by pin-mux setting.

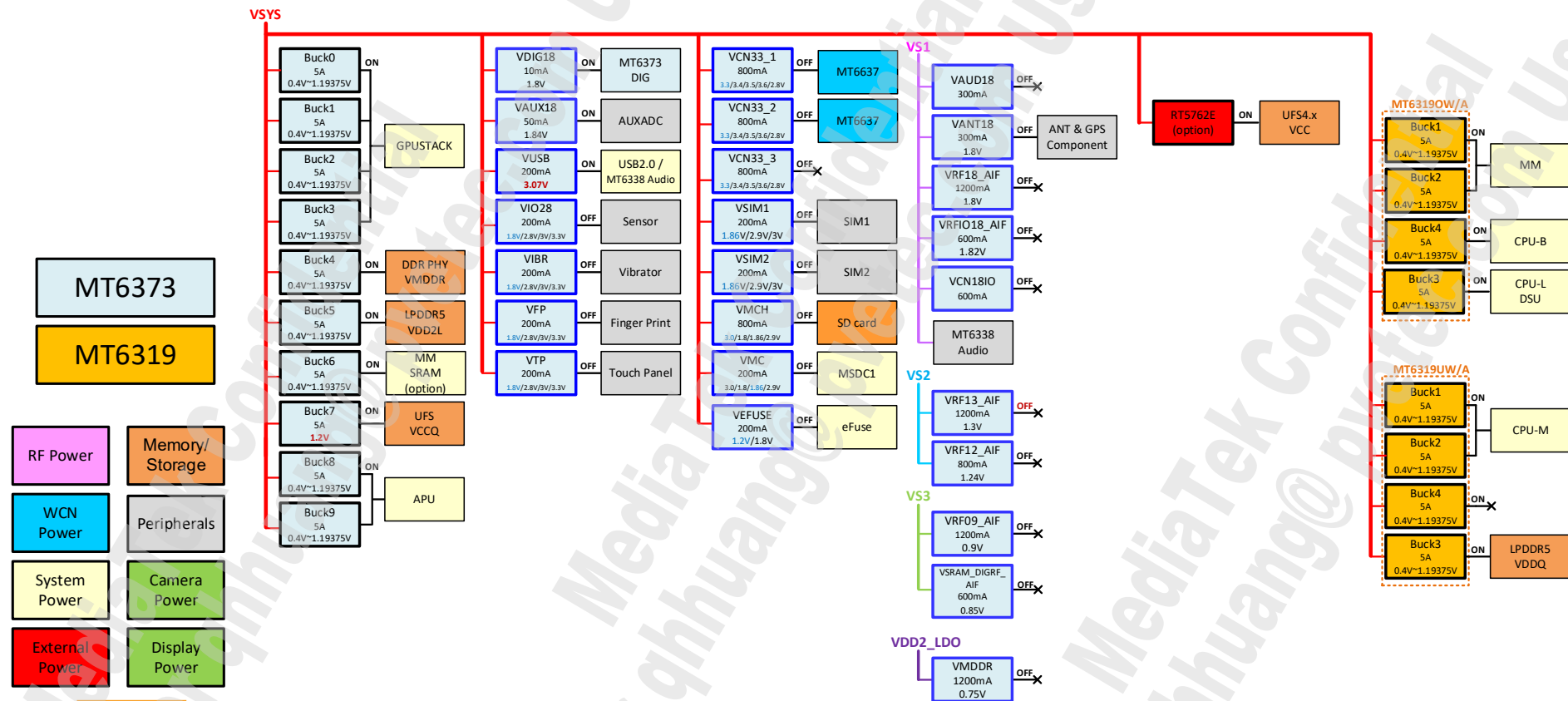
MT8676 Power Grid: MT6363

Note:

1. VRFO18 LDO powers to WCN IC; VCN15 LDO powers to RF IC.



MT8676 Power Grid: MT6373, MT6319*2



MT6363 – Buck Power Plan

Circuit Type	Buck Name (Application name)	Output Voltage (V)	Boot Default (V)	I _{OUT-MAX} (mA)
Buck	BUCK1 (VDIGRF_PMU)	0.4 ~ 1.19 (6.25 mV/step)	ON (0.7)	5,000
	BUCK2/3 (DVDD_MODEM)	0.4 ~ 1.19 (6.25 mV/step)	ON (0.8)	10,000
	BUCK4 (DVDD_SRAM_CORE)	0.4 ~ 1.19 (6.25 mV/step)	ON (0.75)	5,000
	BUCK5 (EMI_VDD2H)	0.4 ~ 1.19 (6.25 mV/step)	ON (1.05)	5,000
	BUCK6/7 (DVDD_CORE)	0.4 ~ 1.19 (6.25 mV/step)	ON (0.75)	10,000
	VS1 (VS1_PMIC)	1.8 ~ 2.2 (12.5 mV/step)	ON (2.0)	2,600
	VS2 (VS2_PMIC)	1.2 ~ 1.5 (12.5 mV/step)	ON (1.45)	4,500
	VS3 (VS3_PMIC)	0.8 ~ 1.19 (6.25 mV/step)	ON (1.05)	4,500

MT6363 – LDO Power Plan (1/4)

Circuit Type	LDO Name	Output Voltage (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Application
VDIG	VDIG18	1.80	ON (1.8)	10	PMIC Interface
ALDO	VAUX18	1.84	ON (1.84)	50	AUXADC
ALDO	VTREF18	1.84	OFF	50	AUXADC pull-up
DLDO	VEMC	2.55/3	ON (2.55)	800	UFS

Note:

- 1) VDIG18, VAUX18, VTREF18 are PMIC internal reference power. DON'T share to other applications.
- 2) Default OFF LDO, voltage is set by software.

MT6363 – LDO Power Plan (2/4)

Circuit Type	LDO Name	Output Voltage (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Application
SLDO	VIO18	1.8	ON (1.8)	600	I/O
SLDO	VA15	1.5	ON (1.5)	600	AP
SLDO	VM18	1.84	ON (1.84)	600	DRAM
SLDO	VRF18	1.84	OFF	600	RF
SLDO	VRFIO18	1.82	OFF	600	Connectivity
SLDO	VCN15	1.82	OFF	300	RF
SLDO	VUFS18	1.86	ON (1.86)	1,200	UART IO

Dedicated for IO,
not for other application

Note:

- 1) VRFIO18 LDO powers to Connectivity IC
- 2) VCN15 LDO powers to RF IC.
- 3) Default OFF LDO, voltage is set by software.

MT6363 – LDO Power Plan (3/4)

Circuit Type	LDO Name	Output Voltage (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Application
SLDO	VA12_1	1.2	ON (1.2)	600	AP
SLDO	VA12_2	1.2	ON (1.2)	600	AP
SLDO	VCN13	1.35	OFF	800	Connectivity
SLDO	VRF13	1.34	OFF	800	RF
SLDO	VRF12	1.24	ON (1.23)	600	RF
SLDO	VUFS12	1.2	ON (1.2)	1200	AP
SLDO	VSRAM_CPUB	0.5~1.1925	ON (0.75)	600	CPU_B SRAM
SLDO	VSRAM_CPUM	0.5~1.1925	ON (0.75)	600	CPU_M SRAM
SLDO	VSRAM_CPUL	0.5~1.1925	ON (0.75)	1,200	CPU_L SRAM

Note:

1) Default OFF LDO, voltage is set by software.

MT6363 – LDO Power Plan (4/4)

Circuit Type	LDO Name	Output Voltage (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Application
SLDO	VRF09	0.9	OFF	800	RF
SLDO	VSRAM_MODEM	0.4~1.1925	ON (0.8)	1,200	MODEM SRAM
SLDO	VSRAM_DIGRF	0.4~1.1925	ON (0.85)	600	DIGRF SRAM
SLDO	VSRAM_MDFE	0.4~1.1925	ON (0.9)	600	GPU SRAM
SLDO	VSRAM_APU	0.4~1.1925	ON (0.75)	600	APU SRAM

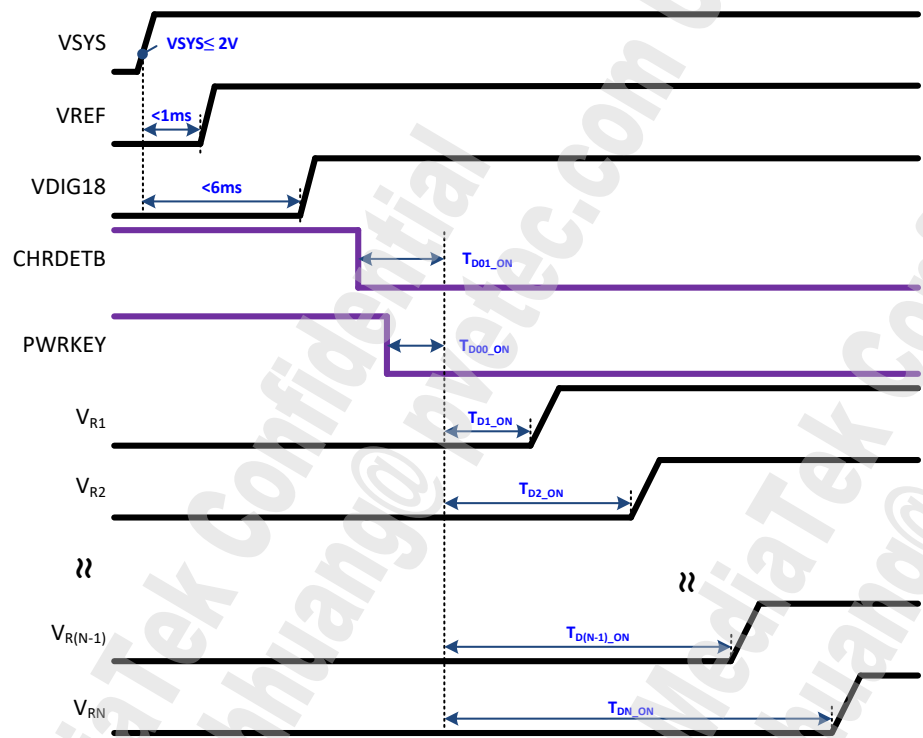
Note:

- 1) Default OFF LDO, voltage is set by software.
- 2) VSRAM_MDFE powers to GPU_SRAM

MT6363 Design Notice

Function Description

MT6363 – Power ON Sequence

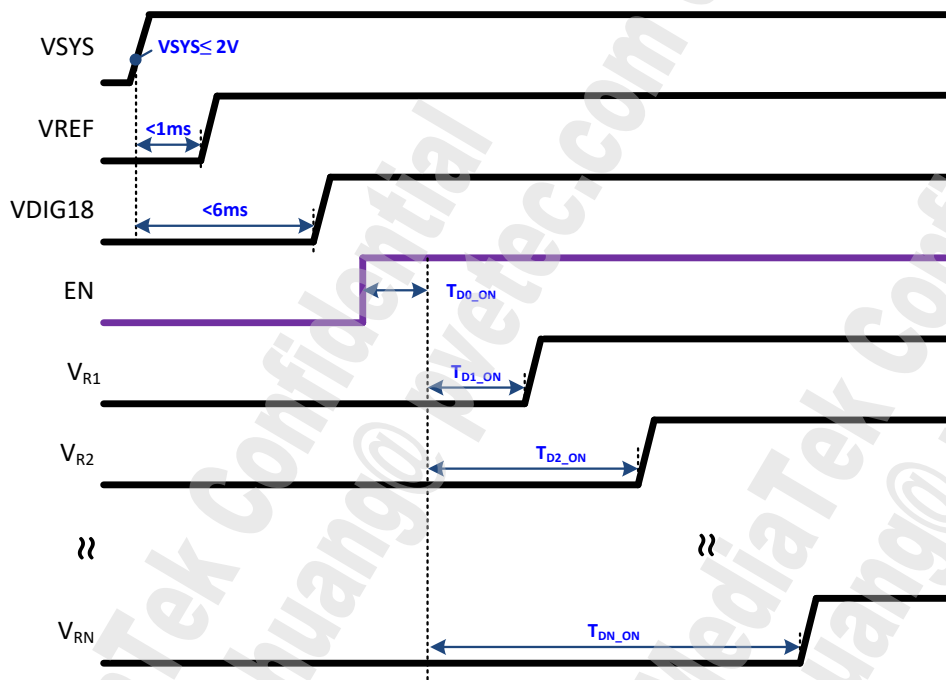


Note:

- 1) The timing parameters $T_{D1_ON} \sim T_{DN_ON}$ are typical values (variation: $\pm 10\%$).
- 2) The timing parameters $T_{D00_ON} \sim T_{D01_ON}$ are typical values (variation: $\pm 20\%$).

T_{DX_ON} (ms)		$PWRKEY = L$ T (ms)	Configuration	Application
T_{D01_ON}	140.0	-	CHRD E TB = L	Start Power ON
T_{D00_ON}	38.0	-	PWRKEY = L	Start Power ON
T_{D1_ON}	10	48	VAUX18	AuxADC
T_{D2_ON}	14	52	VS3	SLDO
T_{D3_ON}	15.5	53.5	BUCK4	VSRAM_CORE
T_{D4_ON}	17	55	VSRAM_APU	APU SRAM
T_{D5_ON}	17	55	VSRAM_MODEM	MODEM SRAM
T_{D6_ON}	18	56	VBUCK6 / VBUCK7	VCORE
T_{D7_ON}	19.5	57.5	VBUCK2 / VBUCK3	VMODEM
T_{D8_ON}	21	59	VBUCK1	VDIGRF
T_{D9_ON}	22.5	60.5	VS2	SLDO
T_{D10_ON}	24	62	VSRAM_MDFE	GPU SRAM
T_{D11_ON}	24	62	VSRAM_DIGRF	RF IC, DIGRF SRAM
T_{D12_ON}	25	63	VS1	SLDO
T_{D13_ON}	26.5	64.5	VIO075	IO
T_{D14_ON}	30.5	68.5	VA12_1	ABB
T_{D15_ON}	30.5	68.5	VA12_2	ABB
T_{D16_ON}	30.5	68.5	VRF12	RF
T_{D17_ON}	30.5	68.5	VUFS12	ABB
T_{D18_ON}	30.5	68.5	EXT_PMIC_EN1	Enable Sub-PMIC
T_{D19_ON}	30.5	68.5	VUFS18	DVDDIO(IOBL)
T_{D20_ON}	39	77	VA15	ABB
T_{D21_ON}	41	79	VIO18	DVDDIO
T_{D22_ON}	43	81	VM18	EMI_VDD1
T_{D23_ON}	44.5	82.5	VEMC	UFS VCC
T_{D24_ON}	44.5	82.5	EXT_PMIC_EN2	Enable Sub-PMIC
T_{D25_ON}	44.5	82.5	VSRAM_CPUL	CPU_L SRAM
T_{D26_ON}	45.5	83.5	VBUCK5	EMI_VDD2H
T_{D27_ON}	53.5	91.5	VSRAM_CPUB	CPU_B SRAM
T_{D28_ON}	53.5	91.5	VSRAM_CPUM	CPU_M SRAM
T_{D29_ON}	70	109.5	SYSRSTB	-

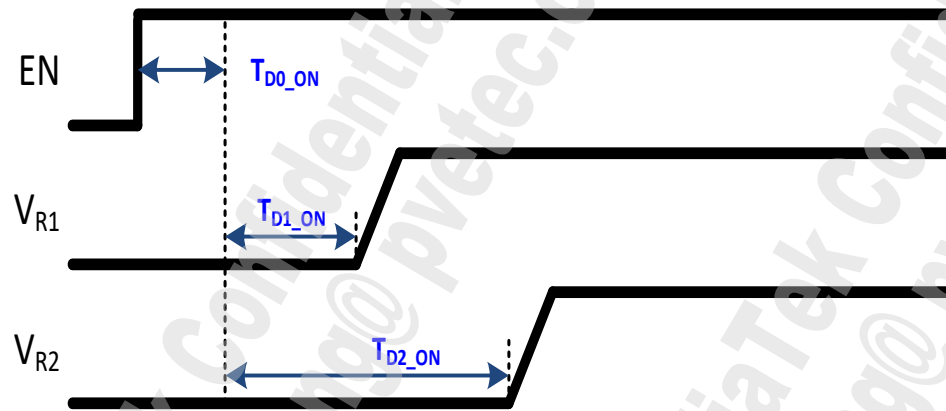
MT6373 – Power ON Sequence



Note: The timing parameters $T_{D0_ON} \sim T_{DN_ON}$ are typical values (variation: $\pm 10\%$).

T_{DX_ON} (ms)		EN=Hi T (ms)	Configuration	Application
T_{D0_ON}	2.15~3.35	-	EN = H (EXT_PMIC_EN1)	enable
T_{D1_ON}	0	3.35	VAUX18	AUXADC
T_{D2_ON}	3	6.35	VBUCK7	UFS VCCQ
T_{D3_ON}	9.5	12.85	VUSB	USB_PHY
T_{D4_ON}	11	14.35	VBUCK6	MM SRAM (option)
T_{D5_ON}	13	16.35	VBUCK8 VBUCK9	APU
T_{D6_ON}	17	20.35	VBUCK5	VDD2L
T_{D7_ON}	17	20.35	VBUCK4	VMDDR
T_{D8_ON}	22	25.35	VBUCK0 VBUCK1 VBUCK2 VBUCK3	GPU_STACK

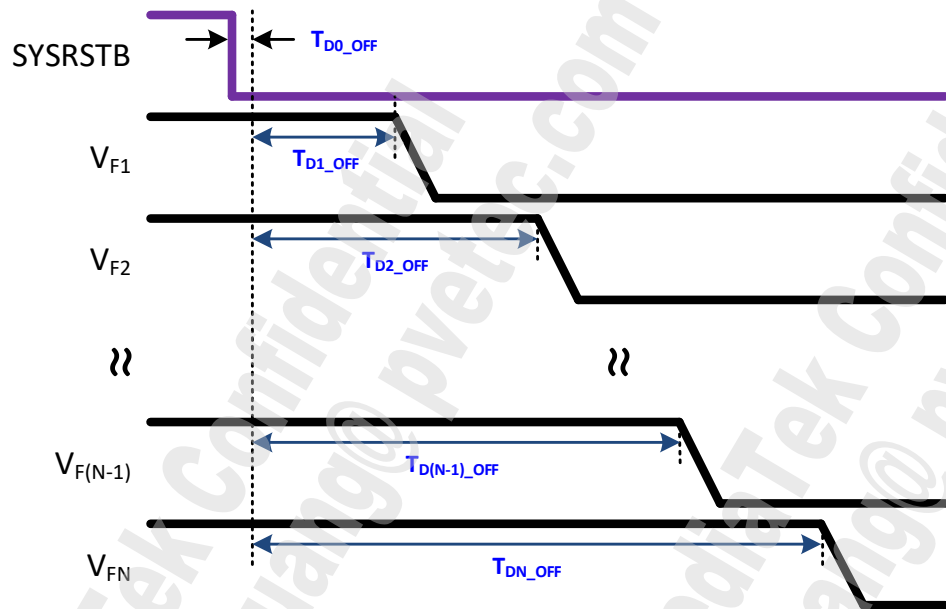
MT6319 – Power ON Sequence



Note: The timing parameters are typical values (variation: $\pm 20\%$).

MT6319 ^{COW}	T (ms)	Configuration	Application
T_{D0_ON}	0.8 ~ 2.0	EN = H (EXT_PMIC_EN2)	Enable
T_{D1_ON}	4	VBUCK3	CPU_L
T_{D2_ON}	6	VBUCK1+2	MM
T_{D3_ON}	12	VBUCK4	CPU_B
MT6319 ^{CUW}	T (ms)	Configuration	Application
T_{D0_ON}	0.8 ~ 2.0	EN = H (EXT_PMIC_EN2)	Enable
T_{D1_ON}	12	VBUCK3	EMI_VDDQ
T_{D2_ON}	14	VBUCK1+2	CPU_M
T_{D3_ON}	16	VBUCK4	No use

MT6363 – Power Off Sequence

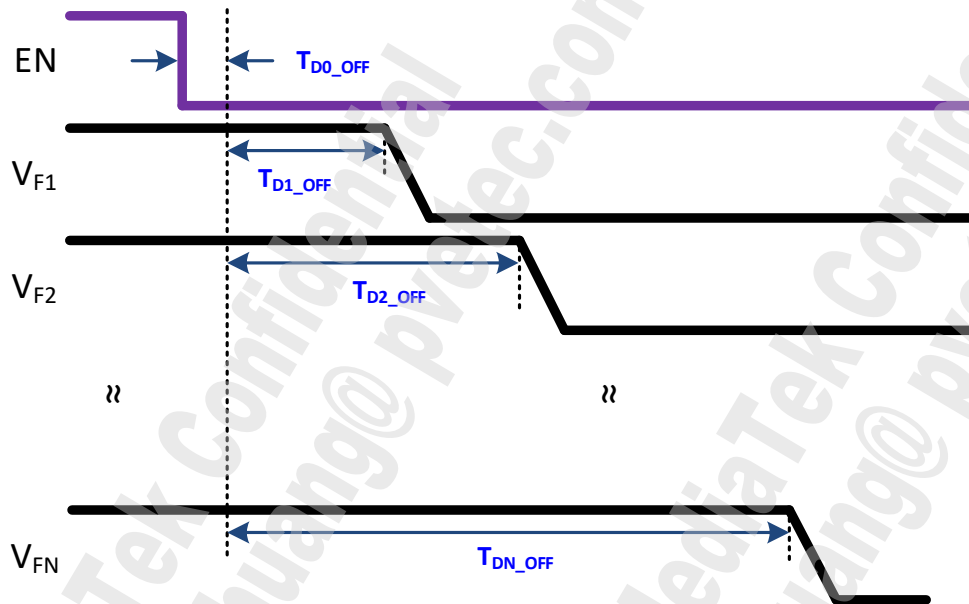


Note:

1. The timing parameters $T_{D1_OFF} \sim T_{DN_OFF}$ are typical values (variation: $\pm 10\%$).
2. Power-OFF sequence starts by SYSRSTB = Hi to Lo.
3. VEMC/VAUX18 can be OFF by software driver control before SYSRSTB = L.

T_{DX_OFF} (ms)		Configuration	Application
T_{D0_OFF}	0	SYSRSTB = L	Start Power OFF
T_{D1_OFF}	2	EXT_PMIC_EN2	Disable Sub-PMIC
T_{D2_OFF}	10	EXT_PMIC_EN1	Disable Sub-PMIC
T_{D3_OFF}	10	VSRAM_CPUB	CPU_B SRAM
T_{D4_OFF}	10	VSRAM_CPUM	CPU_M SRAM
T_{D5_OFF}	24	VSRAM_CPUL	CPU_L SRAM
T_{D6_OFF}	26	VBUCK5	EMI_VDD2H
T_{D7_OFF}	30	VM18	EMI_VDD1
T_{D8_OFF}	30	VEMC	UFS_VCC
T_{D9_OFF}	32	VIO18	IO
T_{D10_OFF}	34	VA15	ABB
T_{D11_OFF}	68	VUFS18	DVDDIO(IOBL)
T_{D12_OFF}	68	VUFS12	ABB
T_{D13_OFF}	68	VA12_1	ABB
T_{D14_OFF}	68	VA12_2	ABB
T_{D15_OFF}	68	VRF12	RF
T_{D16_OFF}	70	VIO075	IO
T_{D17_OFF}	76	VS1	SLDO
T_{D18_OFF}	80	VSRAM_MDFE	GPU SRAM
T_{D19_OFF}	80	VSRAM_DIGRF	DIGRF SRAM
T_{D20_OFF}	82	VS2	SLDO
T_{D21_OFF}	86	VBUCK1	VDIGRF
T_{D22_OFF}	90	VBUCK2 / VBUCK3	VMODEM
T_{D23_OFF}	94	VBUCK6 / VBUCK7	VCORE
T_{D24_OFF}	98	VSRAM_APU	APU SRAM
T_{D25_OFF}	98	VSRAM_MODEM	MODEM SRAM
T_{D26_OFF}	102	VBUCK4	VSRAM_CORE
T_{D27_OFF}	106	VS3	SLDO
T_{D28_OFF}	106	VAUX18	AuxADC

MT6373 – Power Off Sequence

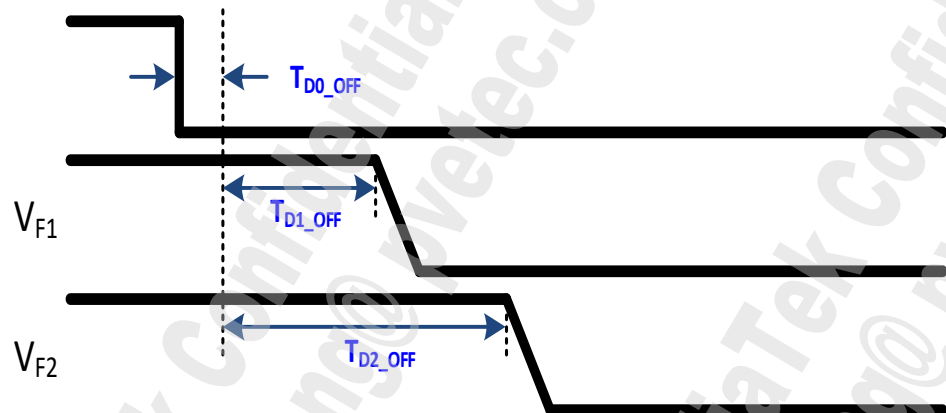


Note:

1. The timing parameters $T_{D1_OFF} \sim T_{DN_OFF}$ are typical values (variation: $\pm 10\%$).
2. Power-OFF sequence is completed after $SYSRSTB = L$.

T_{DX_OFF} (ms)	Configuration	Application
T_{D0_OFF}	< 0.2 EN = L (EXT_PMIC_EN1)	Disable
T_{D1_OFF}	0 VBUCK4	VMDDR
T_{D2_OFF}	4 VBUCK8 VBUCK9	APU
T_{D3_OFF}	4 VBUCK5	VDD2L
T_{D4_OFF}	8 VBUCK6	MM SRAM (option)
T_{D5_OFF}	12 VBUCK0 VBUCK1 VBUCK2 VBUCK3	GPU_STACK
T_{D6_OFF}	16 VUSB	USB_PHY
T_{D7_OFF}	24 VAUX18	AUXADC
T_{D8_OFF}	40 VBUCK7	UFS VCCQ

MT6319 – Power Off Sequence



Note:

1. The timing parameters are typical values (variation: $\pm 20\%$).
2. Power-OFF sequence is completed after $SYSRSTB=L$
3. MM can be OFF by software driver control before $SYSRSTB=L$.

MT6319COW	T (ms)	Configuration	Application
T_{D0_OFF}	<0.2	EN = L (EXT_PMIC_EN2)	Disable
T_{D1_OFF}	0	VBUCK1+2	MM
T_{D2_OFF}	4	VBUCK4	CPU_B
T_{D3_OFF}	12	VBUCK3	CPU_L
MT6319CUW	T (ms)	Configuration	Application
T_{D0_OFF}	<0.2	EN = L (EXT_PMIC_EN2)	Disable
T_{D1_OFF}	0	VBUCK3	EMI_VDDQ
T_{D2_OFF}	4	VBUCK1+2	CPU_M
T_{D3_OFF}	8	VBUCK4	No use

MT6363 – UVLO (Under Voltage Lock Out)

MT6363 UVLO		
Sensing pin	VSYSNS	VBBSNS
UVLO high threshold	2.6V ~ 3.2V (0.1V/step) Default: 2.9V	2.6V ~ 3.2V (0.1V/step) Default: 2.9V
UVLO low threshold	2.0V ~ 2.9V (0.1V/step) Default: 2.5V	2.5V ~ 2.9V (0.1V/step) Default: 2.6V
Accuracy	± 50 mV	± 50 mV

Note:

1. If UVLO_OFF is set < 2.5V, RG_UVLO2V0_DDLO_EN must be set to 1.

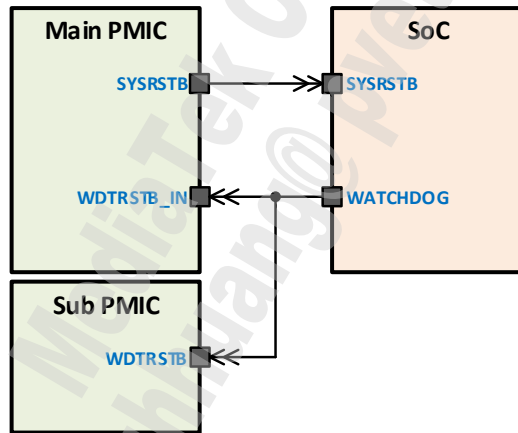
MT6363 – System Reset

• SYSRSTB Function

- SYSRSTB is MT6363's output only.
- Reset to the default value and force the SoC restoration
 - SYSRSTB=Lo to Hi: MT6363 completes power-on sequence.
 - There are two events (PWRKEY=Lo & CHRDET=Lo) for MT6363 power-on sequence start.
 - SYSRSTB=Hi to Lo: MT6363 power-off or exception off
- SYSRSTB pin output driving capability selection.
 - 7.5 mA (default)

• Layout notice

- SYSRSTB and WDTRSTB_IN need to be kept far from noise traces.
- If possible, add "GND" shielding to avoid noise coupling.

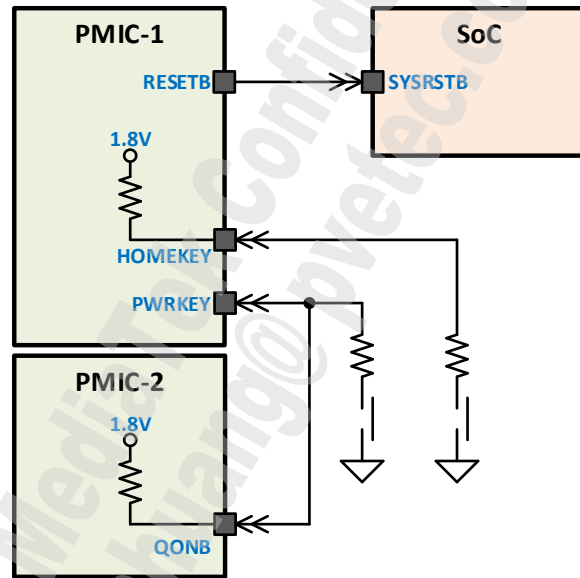


MT6363 – CHRDET B

- **CHRDET B**
 - There are two trigger events (PWRKEY=Lo & CHRDET B=Lo) for MT6363 power-on sequence.
- **CHRDET B**
 - Internal pull-up 40K Ω to VBBSNS
 - $V_{IH} \geq VBBSNS * 70\%$
 - $V_{IL} \leq VBBSNS * 30\%$
 - CHRDET B response time to power-on: 138ms $\pm$ 20%

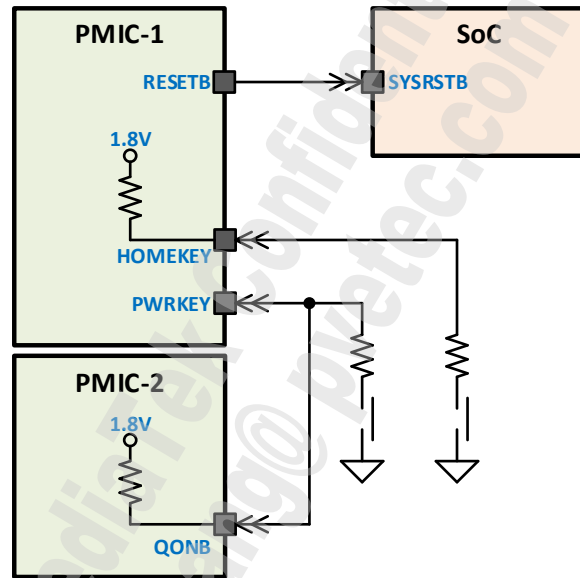
MT6363 – PWRKEY

- **PWRKEY**
 - There are two trigger events (PWRKEY=Lo & CHRDET=Lo) for MT6363 power-on sequence.
- **PWRKEY**
 - Interrupt for press and release key
 - Multi-Keys combination with HOMEKEY for long press shutdown
 - Default: PWRKEY 8s long press shutdown
 - $V_{IH} \geq 1.45V$
 - $V_{IL} \leq 0.3V$
 - PWRKEY response time to power-on: $35ms \pm 20\%$



MT6363 – PWRKEY Long Press Shutdown

- **PWRKEY long press shutdown**
 - Timer spec.: 2/5/8 (default)/11 sec $\pm 10\%$
 - Timer can be set by software. (CLK source = internal 32 kHz)
 - Key combination is selected by register.
- **Long press shutdown: Multi-Keys combination**
 - PWRKEY (default)
 - PWRKEY + HOMEKEY
 - PWRKEY + HOMEKEY-2
 - PWRKEY + HOMEKEY + HOMEKEY-2
- **PWRKEY long press function**
 - Force a PMIC shutdown. (default)
 - All power supplies are turned off except VDIG18.
 - Disable PWRKEY long press shutdown function
 - JUST change to others key combination and don't let HOMEKEY/HOMEKEY-2 pull-down.



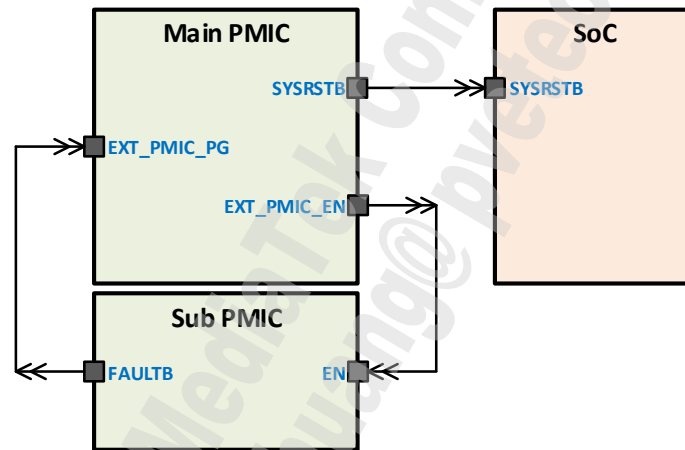
MT6363 – HOMEKEY, HOMEKEY2

- **HOMEKEY, HOMEKEY2 function**
 - Interrupt for press and release key
 - Multi-Keys combination with PWRKEY for long press shutdown
 - $V_{IH}: \geq 0.7*(DVDD18_IO)$
 - $V_{IL}: \leq 0.3*(DVDD18_IO)$
 - HOMEKEY/ HOMEKEY2 response time to interrupt: 34ms $\pm$ 10%

Ball name	Function
GPIO8	HOMEKEY
GPIO9	HOMEKEY2

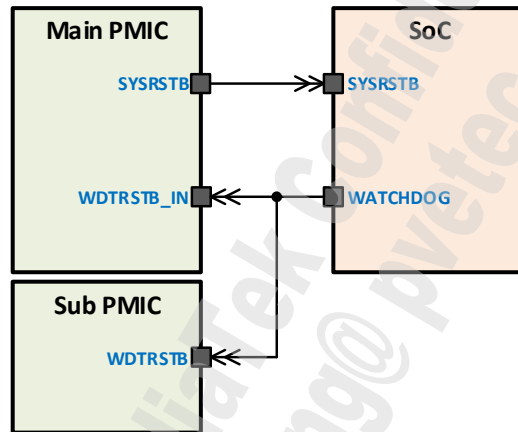
MT6363 – Others PMIC Exception Alarm

- **To enable sub PMIC exception alarms:**
 - MT6363's EXT_PMIC_PG should be connected to **FAULTB** of sub PMIC.
 - MT6363's EXT_PMIC_PG should be active low.
 - MT6363's EXT_PMIC_PG has internal pull-up resistor of 40 kΩ to VSYS.
 - $V_{IH}: \geq 0.7 \cdot V_{SYS}$
 - $V_{IL}: \leq 0.3 \cdot V_{SYS}$
- **Sub PMIC exception conditions**
 - UVLO off
 - OCP
 - Thermal shutdown
 - Sub PMIC VR power is not good.



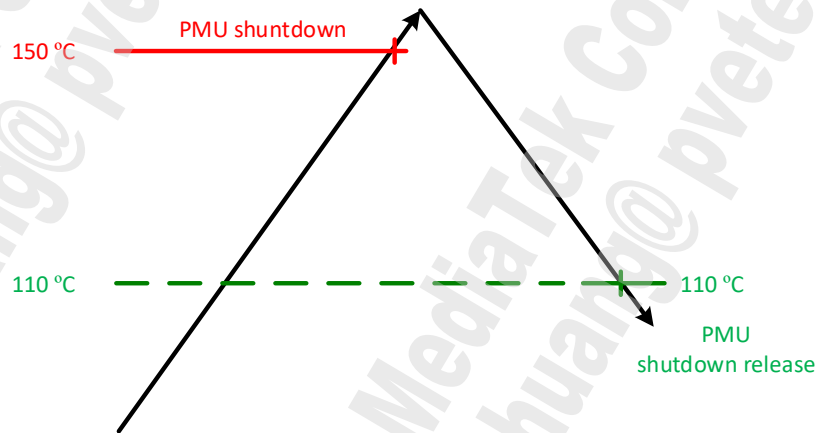
MT6363 – SoC Watchdog Reset

- **SW watchdog reset function (default: off)**
 - MT6363 **WDTRSTB_IN** is input pin and should be connected to SoC **WATCHDOG**.
 - If the SoC watchdog timeout is triggered, SoC will assert **WATCHDOG** to low.
 - The PMICs will be reset if **WDTRSTB_IN** is logic low.
- **Watchdog reset option**
 - RG reset: Reset PMIC register value. **(default)**
 - Cold reset: Shut down PMIC and then watchdog will automatically re-power on.
- **HW PMIC reset function**
 - Response time: 1.7 ms
 - Power source is VIO18_PMU.



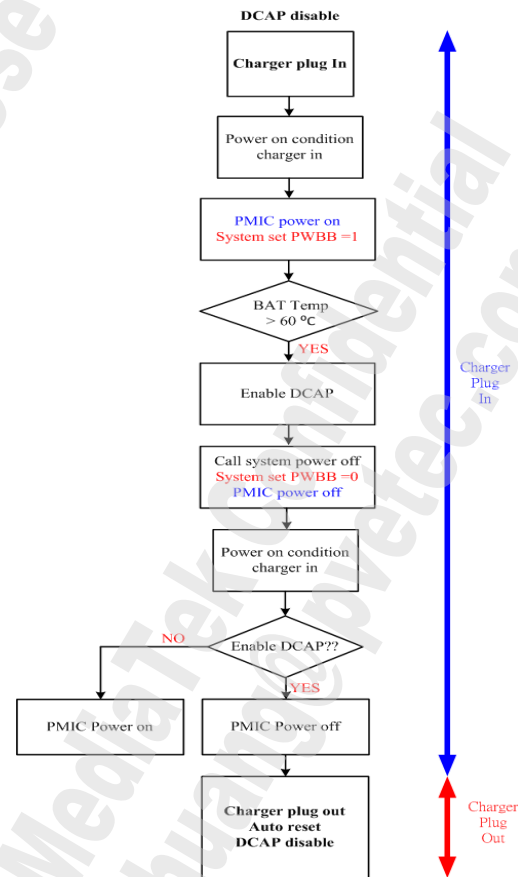
MT6363 – Thermal Shutdown

- **HW control ($\pm 10^{\circ}\text{C}$)**
 - $T_j > 150^{\circ}\text{C}$ PMU HW shutdown
 - HW shutdown release when $T_j < 110^{\circ}\text{C}$
 - Pull PWRKEY = L for next power-on sequence.



MT6363 – DCAP

- **Disable the charger in auto power on (DCAP) condition when the temperature is high.**
 - Enable flow
 - Enable DCAP by RG setting when the charger is plugged in and BAT temp. > HT shutdown level (default: 60°C).
 - Release flow
 - DCAP will be disabled automatically when the charger is unplugged.
 - Notice
 - Enable DCAP function
 - Via MT6363 setting
 - MT6375's CHRDETb keep Low w/o WATCHDOG reset while charger in exist.



✘ DCAP enable flow needs to follow the programming guide.

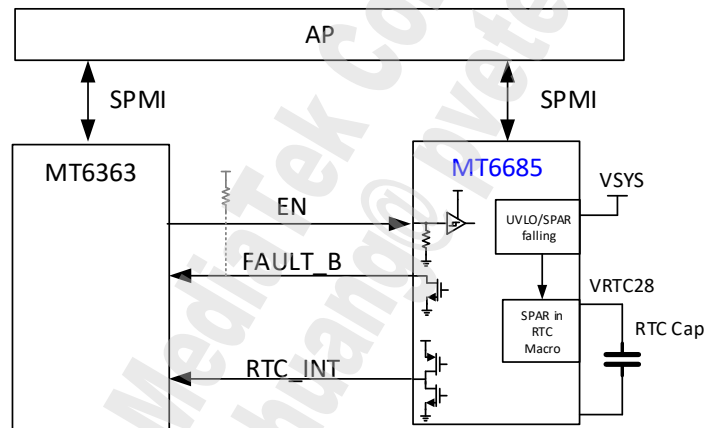
SPAR

- **Feature support condition**

- This feature is a power reboot function to handle sudden battery power drop event.
- SPAR function is triggered by MT6685 where the RTC is designed in.
- Once SPAR occurs, RTC_INT will be pulled high to reboot MT6363.

- **Enabled by software**

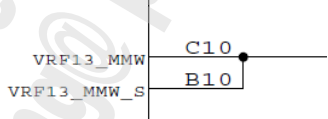
- There are three time settings can be selected.
 - 0.3s, VRTC output needs coin cell or capacitor of > 2.2 uF
 - 0.86s, VRTC output needs coin cell or capacitor of > 4.7 uF
 - 1.83s, VRTC output needs coin cell or capacitor of > 22 uF
 - Min. SPAR trigger condition: "duration of (VSYS < UVLO)" > 1 ms



MT6363 – I/O Power Domain

Pin name	Power source – ball name	Power source domain
PWRKEY	VBBSNS	VSYS
HOMEKEY, HOMEKEY2	DVDD18_IO	VIO18 (1.8V)
CHRDETB	VBBSNS	VSYS
SYSRSTB	VIO18	VIO18 (1.8V)
EXT_PMIC_EN	VSYSNS	VSYS
EXT_PMIC_PG	VBBSNS	VSYS
WDTRSTB_IN	DVDD18_IO	VIO18 (1.8V)
SRCLKEN_IN0, SRCLKEN_IN1	DVDD18_IO	VIO18 (1.8V)
EINT	DVDD18_IO	VIO18 (1.8V)
RTC_INT	VSYSNS	VSYS

MT6363 – Un-used Pin Configuration

Pin name	Description	If un-used
EXT_PMIC_EN1 EXT_PMIC_EN2	Ext. PMIC enable pin	Floating
EXT_PMIC_PG	Ext. PMIC PG pin	Floating
HOMEKEY HOMEKEY2	Key	Floating
RTC_INT	RTC interrupt for SPAR power-on	Floating
AUXADC_VINx	AuxADC	GND
LDO_XX	Default OFF un-used LDOs	w/ remote sense: config as shown 
		w/o remote sense: Floating

Note:

- Please check HOMEKEY/HOMEKEY-2 connection w/ PWRKEY long press shutdown function.

MT6363 – HW Trapping for VEMC Boot Voltage

- **UFS VCC 2.5V or 3V**
 - VEMC LDO with HW trapping design in MT6363 for UFS2.x VCC=3V & UFS3.x VCC=2.5V

Ball: EINT	VEMC boot voltage
Pull low to GND	2.55V
Pull high to VIO18	3V

MT6363 – SPMI I/F

SPMI

- The SPMI is a two-wire interface:
 - 1 serial bi-directional data signal (SDATA)
 - 1 clock signal (SCLK) controlled by master
- SPMI allows the use of up to PMIC*8 pcs on a single bus.
- The power source of SPMI I/O is DVDD18_IO.
- MT6363 supports up to 26 MHz for SPMI speed with load up to 50 pF.

Layout Constraint of SPMI

- SPMI_SCL and SPMI_SDA layout traces should be differential and use GND shielding to prevent noise coupling.
(e.g., VBUS, BUCK_LX, SWCHR_LX, BUCK input pin)
- The traces should not overlap with noise sources on layers N+1 and N-1.
(e.g. VBUS, BUCK_LX, SWCHR_LX, BUCK input pin)
- The max. length of SPMI bus between SoC and PMIC should be less than 6 inches.

MT6363 – AUXADC

- GPIO1~GPIO7(AUXADC_VIN1~AUXADC_VIN7)
 - The power source of AUXADC_VINx is VAUX18.
 - There are three pull-up voltage options:
 - Internal pull-up to VAUX18: internal pull-up resistor (30 k Ω / 100 k Ω / 400 k Ω).
 - The capacitance at AUXADC_VINx should be NC.
 - External pull-up to VTREF18 (**pull-up resistor must be higher than 10k Ω**)
 - The capacitance at AUXADC_VINx should be less than 1 nF and it should be put near the PMIC side.
 - External pull-up to VIO18 (**pull-up resistor must be higher than 10k Ω**)
 - The capacitance at AUXADC_VINx should be less than 1 uF and it should be put near the PMIC side.

Ball name	AUXADC_VINx	Application
GPIO1	AUXADC_VIN1	General purpose
GPIO2	AUXADC_VIN2	
GPIO3	AUXADC_VIN3	
GPIO4	AUXADC_VIN4	
GPIO5	AUXADC_VIN5	
GPIO6	AUXADC_VIN6	
GPIO7	AUXADC_VIN7	

Bias: VIO075

- VIO075 is bias circuit, not LDO output.

Circuit type	IP name	Output voltage (V)	Boot default (V)	Application
VBIASN	VIO075	0.75	ON (0.75)	I/O

MT6363 Design Notice

Function Description – DC-DC

DC-DC: Buck Converter List

MT6363 buck (Application)	Output de-coupling (min.)	Output de-coupling (max.)	Total C _{OUT} value (after de-rating)
VS1 (VS1_PMIC)	0.47 uH 10 uF*3	0.47 uH 22 uF*2	≥ 15.6 uF #1
VS2 (VS2_PMIC)	0.47 uH 10 uF*3	0.47 uH 22 uF*2	≥ 15.6 uF #1
VS3 (VS3_PMIC)	0.47 uH 10 uF*3	0.47 uH 22 uF*2	≥ 15.6 uF #1
BUCK1 (VDIGRF_PMU)	0.24 uH 10uF*2	0.24 uH 22 uF*2	≥ 10.3 uF #1
BUCK2 + BUCK3 (DVDD_MODEM)	0.24 uH*2 10 uF*6	0.24 uH*2 22 uF*4	≥ 30.9 uF #1
BUCK4 (DVDD_SRAM_CORE)	0.47 uH 10uF*3	0.47 uH 22 uF*2	≥ 15.6 uF #1
BUCK5 (EMI_VDD2H)	0.24 uH 10uF*12	0.24 uH 22 uF*7	≥ 61.8 uF #1
BUCK6 + BUCK7 (DVDD_CORE)	0.24 uH*2 10uF*6	0.24 uH*2 22uF*4	≥ 30.9 uF #1

Note #1: Total C_{OUT} value @1V derating for AC 0.01Vrms and overall operating temperature.

Note: 1. Cap replacement plan: 10uF/0402/6.3V/X5R *2ea <-> 22uF/0603/6.3V/X5R *1ea
 2. There is no output current margin for peripheral application in VS1 & VS2 power domains.
 3. Add external buck for VS1 or VS2 peripheral application if needed.
 4. Please check the low power mode setting if peripheral application at VS1 or VIO18

DC-DC: Power Inductor Selection Guideline

- For inductors, we recommend following the notice below.
 - Follow the table below for inductor current rating.
 - Small size and high efficiency are our main objectives. The proposed low core loss and low DCR (copper wire resistance) inductors are more efficient. For efficiency test conditions, refer to PMIC datasheet.

MT6363 buck (Application)	Inductance (uH) MUST-BE	I _{SAT,MAX} (mA) MUST-BE
VS1	0.47	≥ 3099
VS2	0.47	≥ 4858
VS3	0.47	≥ 2487
BUCK1 (VDIGRF_PMU)	0.24	≥ 2029
BUCK2 + BUCK3 (DVDD_MODEM)	0.24*2	≥ 4974
BUCK4 (DVDD_SRAM_CORE)	0.47	≥ 2533
BUCK5 (EMI_VDD2H)	0.24	≥ 5471
BUCK6 + BUCK7 (DVDD_CORE)	0.24*2	≥ 5909

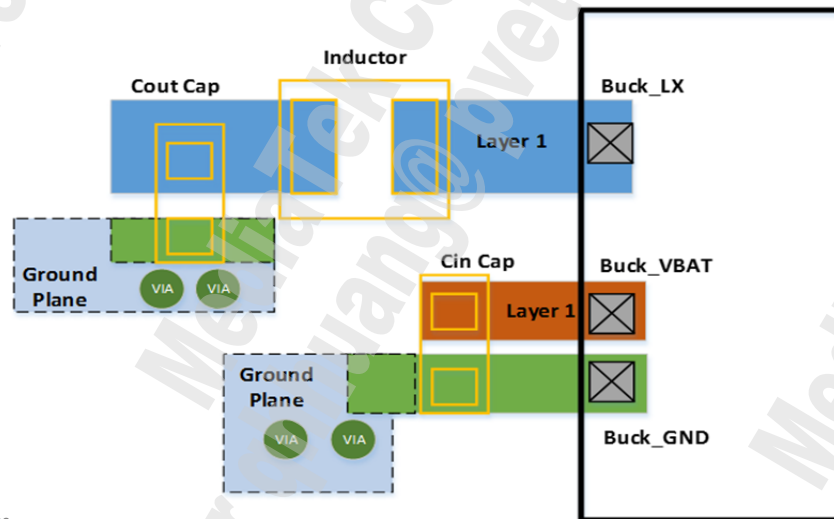
Note:

- The I_{SAT} depends on inductance saturation. (-30% reduction from initial L value)
- The I_{TEMP} depends on ambient temperature (T_A), and T_j should not exceed 125°C.

DC-DC: Schematic and Layout Notice #1

- **DC-DC layout guide #1**

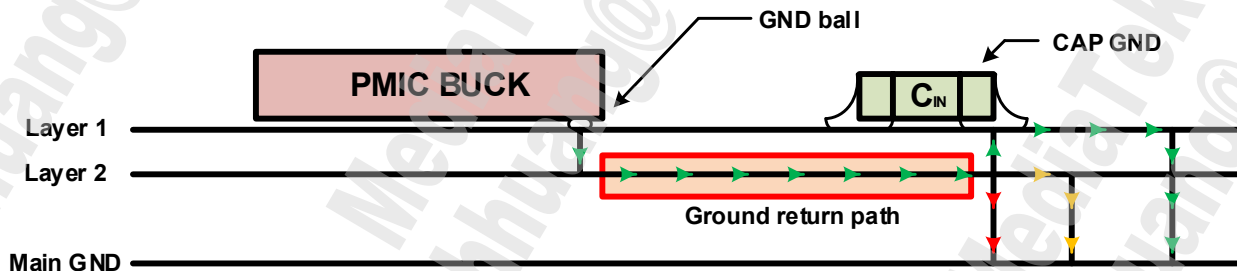
- The concept of buck layout is as shown below.
 - All the input decoupling capacitors should be placed near PMIC.
 - Buck GND balls should be connected to input capacitors firstly, and isolated from the nearby GND trace then connected to main GND plane.
 - The buck inductors should be placed near PMIC.



DC-DC: Schematic and Layout Notice #2-1

- **DC-DC layout guide #2-1: Input capacitor**

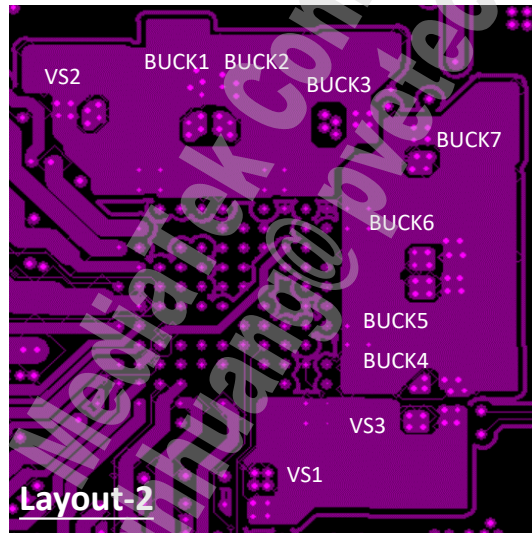
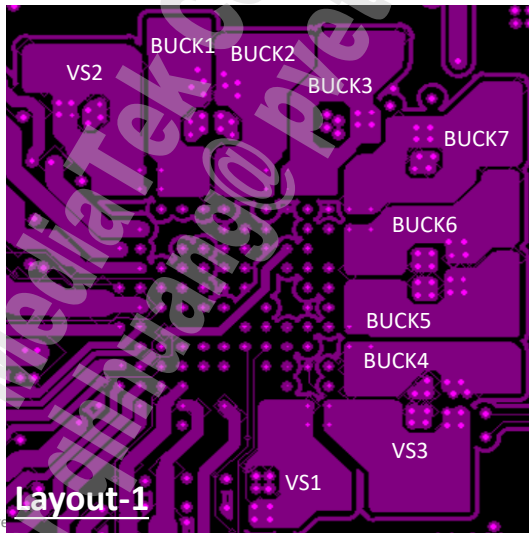
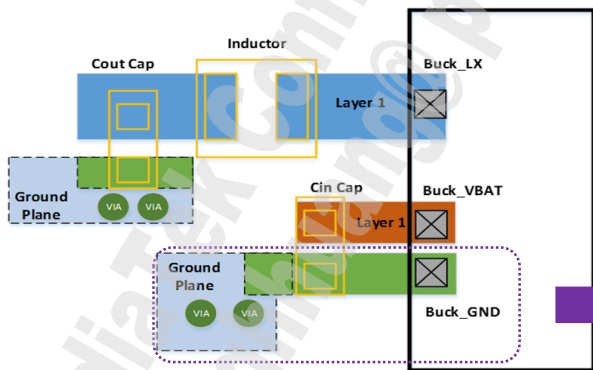
- The concept of buck layout is as shown below.
- Keep ground return path short. Stubs are not allowed.
- To connect buck ground to main ground, you can select the green, yellow, or red path.
 - The green path is recommended. Ground returns from PMIC to input capacitor first, and then connects to the main ground.
 - The yellow or red path is also acceptable, but note that only diverging ground paths underneath input capacitor ground are allowed.
- For more layout constraint details, refer to “MT6985 SCH and PCB Design Check List”.



DC-DC: Schematic and Layout Notice #2-2

• DC-DC layout guide #2-2: Input capacitor GND

- Layout-1 & Layout-2 example can be implement for MT6363 buck input GND.
- Layout-2:
 - GND_VS2/GND_BUCK1/GND_BUCK2/GND_BUCK3 can be one group.
 - GND_BUCK4/GND_BUCK5/GND_BUCK6/GND_BUCK7 can be one group.
 - GND_VS1/GND_VS3 can be one group.



DC-DC: Schematic and Layout Notice #3-1

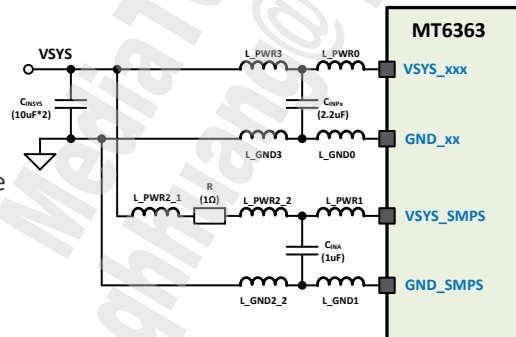
$C_{IN_{SYS}}$ & $C_{IN_{PX}}$ must follow ref. design SCH and below constraint strictly, reduction may cause damaged issue.

• DC-DC layout guide #3-1: Input capacitor (10uF *2 + 2.2uF *10)

MT6363	Input trace	Inductance (from $C_{IN_{SYS}}$ to PMIC ball)	Inductance (from $C_{IN_{PX}}$ to PMIC ball)
BUCK1 BUCK4 BUCK7	VSYS_BUCKx GND_BUCKx	10uF *2 ($L_{PWR3} + L_{GND3} + L_{PWR0} + L_{GND0}$) < 2.4 nH	$C_{IN_{PX}}=2.2uF/0201$ ($L_{PWR0} + L_{GND0}$) < 1 nH
BUCK2 BUCK3 BUCK5 BUCK6	VSYS_BUCKx GND_BUCKx	10uF *2 ($L_{PWR3} + L_{GND3} + L_{PWR0} + L_{GND0}$) < 2.1 nH	$C_{IN_{PX}}=2.2uF/0201$ ($L_{PWR0} + L_{GND0}$) < 1 nH
VS1 VS2 VS3	VSYS_VSx GND_VSx	10uF *2 ($L_{PWR3} + L_{GND3} + L_{PWR0} + L_{GND0}$) < 2.7 nH	$C_{IN_{PX}}=2.2uF/0201$ ($L_{PWR0} + L_{GND0}$) < 2 nH
Controller	VSYS_SMPS GND_SMPS	($L_{PWR2_1} + L_{PWR2_2} + L_{GND2_2} + L_{PWR1} + L_{GND1}$) < 11 nH	($L_{PWR1} + L_{GND1}$) < 3 nH

Note #1: Effective $C_{IN_{PX}}$ (2.2 uF) value ≥ 0.36 uF @5V derating for AC 0.2 Vrms and overall operating temperature

Note #2: $C_{IN_{SYS}} = 10$ uF/0402/6.3V/X5R*2

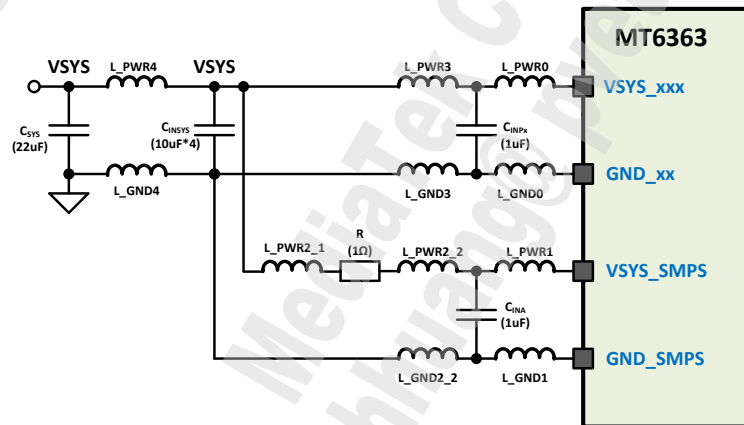


DC-DC: Schematic and Layout Notice #3-2

C_{INSYS} & C_{INPX} must follow ref. design SCH and below constraint strictly, reduction may cause damaged issue.

- DC-DC layout guide #3-2: Input capacitor ($10\mu F * 4 + 1\mu F * 10$)**

MT6363	Input trace	Inductance (from C_{SYS} to PMIC ball)	Inductance (from C_{INSYS} to PMIC ball)	Inductance (from C_{INPX} to PMIC ball)
BUCK1~7, VS1, VS2, VS3	VSYS_BUCKx GND_BUCKx VSYS_VSx GND_VSx	$22\mu F$ ($L_{PWR4} + L_{GND4} + L_{PWR3}$ + $L_{GND3} + L_{PWR0} + L_{GND0}$) < 4.5 nH	$10\mu F * 4$ ($L_{PWR3} + L_{GND3} + L_{PWR0} + L_{GND0}$) < 2 nH	$C_{INPX}=1\mu F/0201$ ($L_{PWR0} + L_{GND0}$) < 0.6 nH
Controller	VSYS_SMPS GND_SMPS	($L_{PWR2_1} + L_{PWR2_2} + L_{GND2_2} + L_{PWR1} + L_{GND1}$) < 11 nH		($L_{PWR1} + L_{GND1}$) < 3 nH



Note #1: $C_{INPX} = 1\mu F/0201/6.3V/X5R$

Note #2: $C_{INSYS} = 10\mu F/0402/6.3V/X5R*4$

DC-DC: Schematic and Layout Notice #3-3

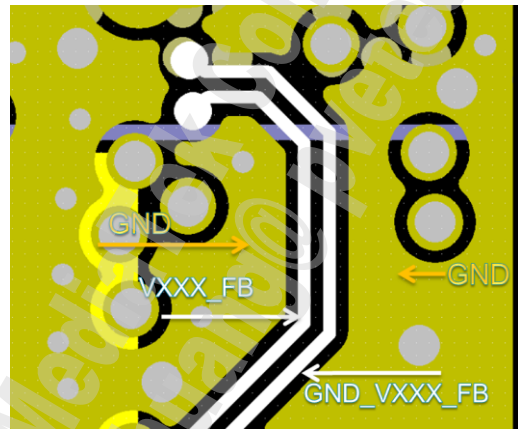
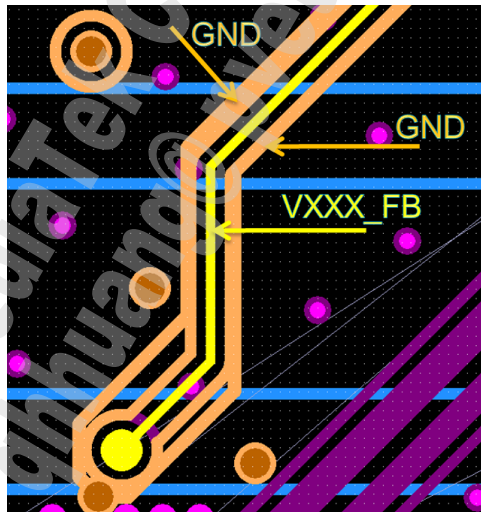
- **DC-DC layout guide #3-3: Input capacitor example**
 - L_PWR1 (VSYS_BUCKx to C_{INPX} VSYS pad): 10-layer (green trace)
 - L_GND1 (GND_BUCKx to C_{INPX} GND pad): 9-layer (purple trace)
 - Smaller L_GND1: larger size of purple trace
 - Smaller L_PWR1: larger overlapping size of purple trace and green trace



DC-DC: Schematic and Layout Notice #4

- **DC-DC layout guide #4: Remote sense**

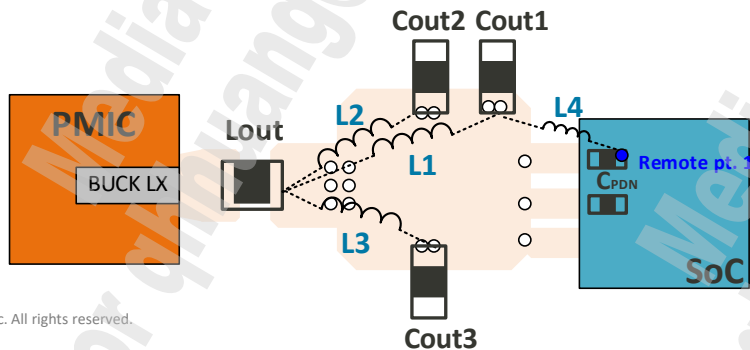
- The feedback trace must be shielded from all noise sources, especially the clock signal and LX nodes of DC/DC converter.
 - BUCK1_FB, GND_BUCK1_FB
 - BUCK2_FB, GND_BUCK2_FB
 - ~~BUCK3_FB, GND_BUCK3_FB~~
 - BUCK4_FB, GND_BUCK4_FB
 - BUCK5_FB, GND_BUCK5_FB
 - BUCK6_FB, GND_BUCK6_FB
 - ~~BUCK7_FB, GND_BUCK7_FB~~



DC-DC: Schematic and Layout Notice #5

• DC-DC layout guide #5: Remote sense

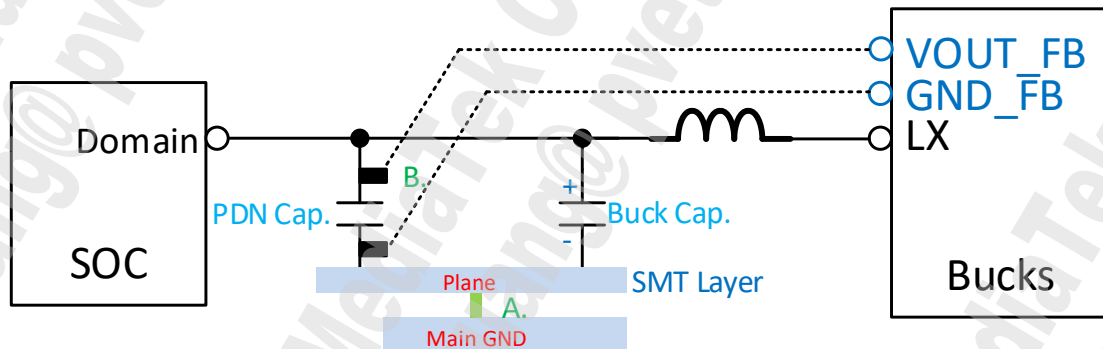
- Find the path with the lowest inductance. The remote point can be located within $C_{outlowest} L + 1 \text{ nH}$.
- The capacitor of remote point needs to be greater than $1 \mu\text{F}$.
- For example, in the case that C_{out1} is a path with the lowest inductance:
 - If inductance between C_{out1} and remote point 1 is $L4$, then $L4$ must be lower than 1 nH ($L4 < 1 \text{ nH}$).
 - The remote point can be located between C_{out1} and remote point 1.
- Same rules are applicable to $L2$ & $L3$ if they are the lowest inductance paths.
- The cases above are for multipath. In case of single path, consider the only path the lowest inductance path directly.
- Inductance between C_{out1} and other C_{out} (C_{out2}/C_{out3}) must be lower than 1.5 nH ($L1 + L2 < 1.5 \text{ nH}/L1 + L3 < 1.5 \text{ nH}$)



DC-DC: Schematic and Layout Notice #6

- DC-DC layout guide #6: Remote sense**

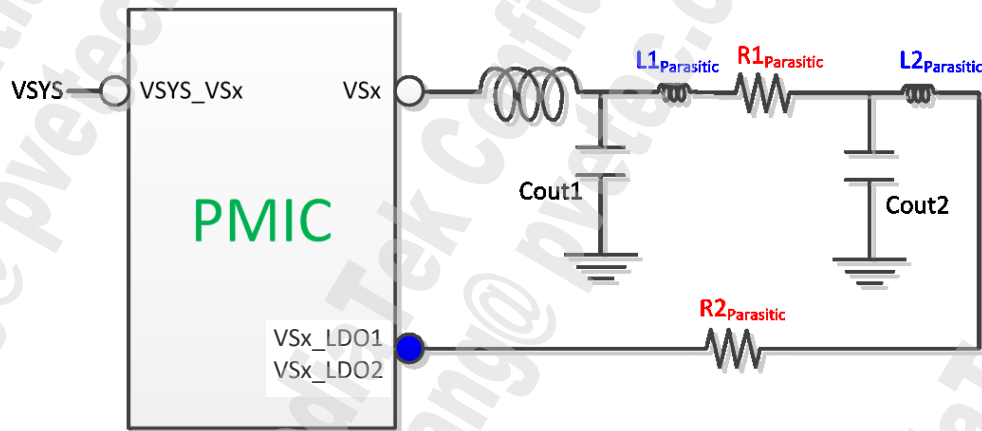
- A. Connect the GNDs of PDN capacitor and buck capacitor on the same layer before connecting them to the main GND. (L8/L9 if SMT is in L10)
- B. GND_FB and VOUT_FB must be connected to PDN capacitor pads directly.
- C. PDN capacitor must be greater than 1 uF.
- D. PDN capacitor must be greater than 4 uF if PDN cap. and buck Cout do not follow rule A and B.



DC-DC: Schematic and Layout Notice #7

DC-DC layout guide #7: Remote sense

- The inductance between any C_{outs} must be lower than 1 nH.



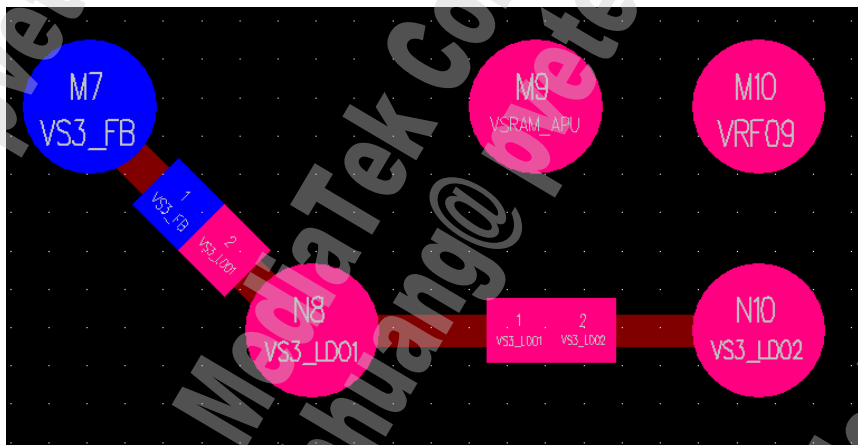
Buck name	LDO input ball (direct-sense to)	Max. R1 + R2 (mΩ)	Max. L1 + L2 (nH)	Max. L1 (nH)
VSx	VSx_LDO2	25	4	1

DC-DC: Schematic and Layout Notice #8

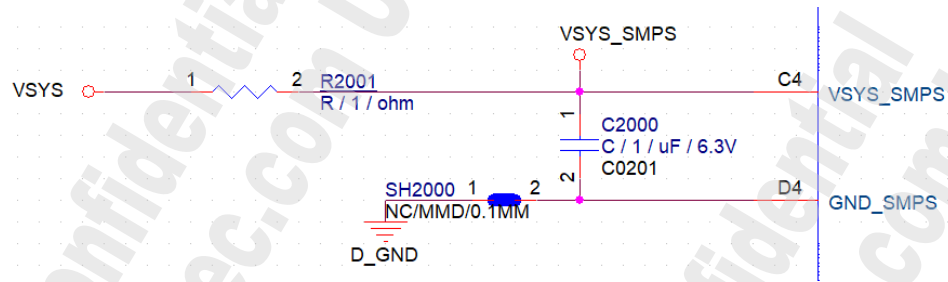
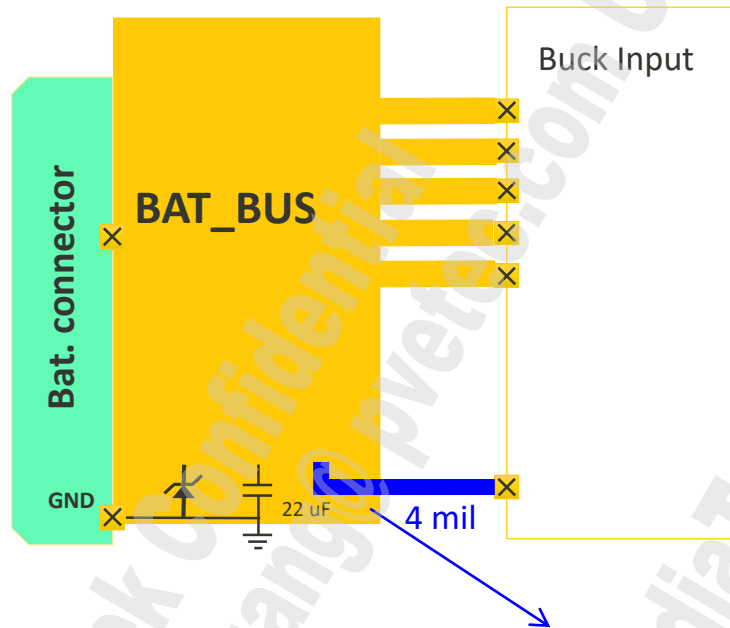
- **DC-DC layout guide #7: Remote sense about VS1, VS2, VS3**

- VS1_FB (ball M5) MUST be connected to VS1_LDO1 (ball N12) and VS1_LDO2 (ball L13) before fan-out power trace.
- VS2_FB (ball C13) MUST be connected to VS2_LDO1 (ball E13) and VS2_LDO2 (ball J13) before fan-out power trace.
- VS3_FB (ball M7) MUST be connected to VS3_LDO1 (ball N8) and VS3_LDO2 (ball N10) before fan-out power trace.

- Layout example:



DC-DC: Schematic and Layout Notice #9

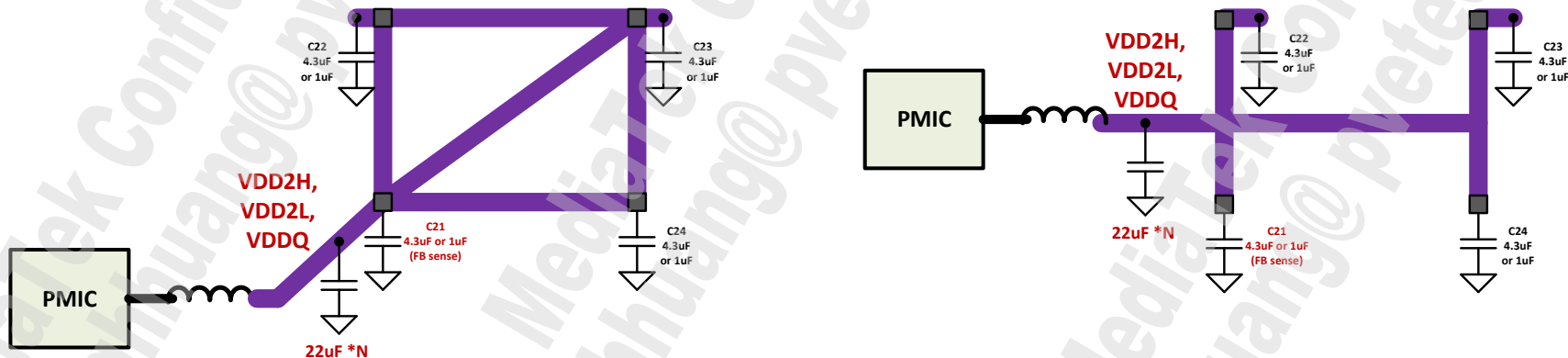


Note:

1. Buck controller power trace (VSYS_SMPS) must be single trace connected to battery VBAT_BUS directly, and it can't be merged with other traces.
2. VSYS adds $1\Omega \pm 5\%$ /0201 resistor to VSYS_SMPS for noise rejection.

DC-DC: Schematic and Layout Notice #10

- **DC-DC layout guide #8: DRAM power capacitor**
 - A. Below DRAM power trace layout example for reference
 - B. The 10uF/22uF output capacitors MUST place between power inductor and DRAM power ball.
 - C. The buck FB sense point (C21) MUST close to 10uF/22uF output capacitors to pass MES.



MT6363 Design Notice

Function Description – LDO

LDO: Input Capacitor Table

Ball name	Input capacitance (Typ.)
VSYS_LDO1	2.2uF / 0201 / 6.3V / X5R
VS1_LDO1, VS1_LDO2	10uF / 0402 / 6.3V / X5R*3 #1
VS2_LDO1, VS2_LDO2	10uF / 0402 / 6.3V / X5R*3 #1
VS3_LDO1, VS3_LDO2	10uF / 0402 / 6.3V / X5R*3 #1

Note: Common parts as 2.2uF/0201, 2.2uF/0402, 10uF/0402 can be used.

LDO: Input Power Table

LDO	Control power ball	Input power ball	LDO	Control power ball	Input power ball
VA15	VSYS_LDO1	VS1_LDO1	VRF13	VSYS_LDO1	VS2_LDO1
VRF18	VSYS_LDO1	VS1_LDO1	VRF12	VSYS_LDO1	VS2_LDO1
VRFIO18	VSYS_LDO1	VS1_LDO1	VUFS12	VSYS_LDO1	VS2_LDO1
VCN15	VSYS_LDO1	VS1_LDO1	VSRAM_CPUB	VSYS_LDO1	VS2_LDO1
VM18	VSYS_LDO1	VS1_LDO1	VSRAM_CPUM	VSYS_LDO1	VS2_LDO1
VIO18	VSYS_LDO1	VS1_LDO2	VSRAM_CPUL	VSYS_LDO1	VS2_LDO2
VUFS18	VSYS_LDO1	VS1_LDO2	VA12_1	VSYS_LDO1	VS2_LDO2
VSRAM_APU	VSYS_LDO1	VS3_LDO1	VA12_2	VSYS_LDO1	VS2_LDO2
VSRAM_MDFE	VSYS_LDO1	VS3_LDO1	VCN13	VSYS_LDO1	VS2_LDO2
VSRAM_DIGRF	VSYS_LDO1	VS3_LDO1	VAUX18	VSYS_LDO1	VSYS_LDO1
VSRAM_MOFEM	VSYS_LDO1	VS3_LDO2	VTREF18	VSYS_LDO1	VSYS_LDO1
VRF09	VSYS_LDO1	VS3_LDO2	VEMC	VSYS_LDO1	VSYS_LDO1

LDO: Output Capacitor Table (1/2)

LDO name	Application	C _{O,min} (uF)	C _{O,max} (uF)	LDO name	Application	C _{O,min} (uF)	C _{O,max} (uF)
VDIG18	I/O	1	1	VA12_1	ABB	1	11
VAUX18	AUXADC	1	2.2	VA12_2	ABB	1	11
VTREF18	VTREF18	1	2.2	VCN13	VCN13	9.4	19.1
VEMC	UFS4.x	20	24.7	VRF13	VRF13	9.4	19.7
VIO18	I/O	10.4	30	VRF12	VRF12	3.2	6.6
VA15	VA15	1	11	VUFS12 (Local Sense)	ABB	4.7 + 4.7	44
VM18	DRAM VDD1	6.9	20.6	VSRAM_CPUB	SRAM_B	4.4	5.81
VRF18	RF 1.8V	9.4	11.6	VSRAM_CPUM	SRAM_M	4.4	5.81
VRFIO18	Consys 1.7V	1	5.7	VSRAM_CPUL	SRAM_L	6.9	9.4
VCN15	RFIO 1.82V	9.4	11.5	VRF09	VRF09	19.4	24.1
VUFS18	IO (IOBL)	4.9	14.1	VSRAM_MODEM	VSRAM_MODEM	6.9	9.4
Dedicated for IO, not for other application				VSRAM_DIGRF	VSRAM_DIGRF	4.4	5.81
				VSRAM_MDFE	VSRAM_GPU	4.4	5.81
				VSRAM_APU	VSRAM_APU	4.4	6.5

Note:

- 1) VRFIO18 LDO powers to Connectivity IC
- 2) VCN15 LDO powers to RF IC.
- 3) The output capacitor can be removed if the LDO is off by default and the loading current is 0 mA

LDO: Output Capacitor Table (2/2)

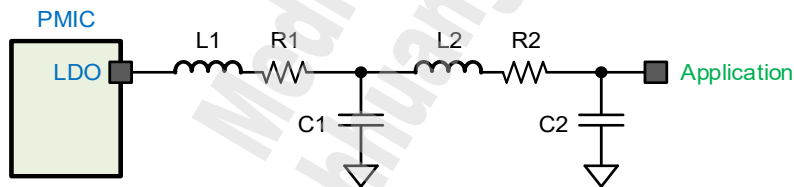
LDO name	Application	PMIC side C _{OUT} (uF)
VTREF18 (default OFF)	Not used	NC
VEMC (default ON)	Enable pin	1

Note : The output capacitor can be removed if the LDO is off by default and the loading current is 0 mA.

LDO: Layout Constraint for LDO Stability (1/3)

LDO layout constraint for stability								
LDO name	$L1 \leq$	$R1 \leq$	$C1_{min}$	$L2 \leq$	$R2 \leq$	$C2_{min}$	$C2_{max}$	$(C1+C2)_{max}$
VIO18	20 nH	0.2Ω	1 uF	20 nH	0.2Ω	9.4 uF	29 uF	30 uF
VA15	20 nH	0.2Ω	1 uF	20 nH	0.2Ω	0 uF	10 uF	11 uF
VM18	7 nH	0.1Ω	2.2 uF	7 nH	0.1Ω	4.7 uF	18.4 uF	20.6 uF
VRF18	20 nH	0.2Ω	4.7 uF	20 nH	0.2Ω	4.7 uF	6.9 uF	11.6 uF
VRFIO18 (for Consys)	20 nH	0.2Ω	1 uF	20 nH	0.2Ω	0 uF	4.7 uF	5.7 uF
VCN15 (for RFIO)	20 nH	0.2Ω	4.7 uF	20 nH	0.2Ω	4.7 uF	6.8 uF	11.5 uF
VUFS18	7 nH	0.2Ω	4.7 uF	20 nH	0.2Ω	0.2 uF	9.4 uF	14.1 uF

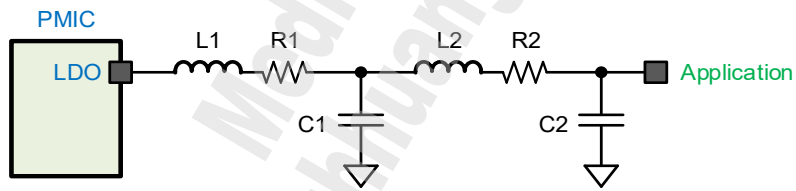
Dedicated for IO,
not for other application



Note: This constraint for LDO stability; application PCB IR should be considered by another check.

LDO: Layout Constraint for LDO Stability (2/3)

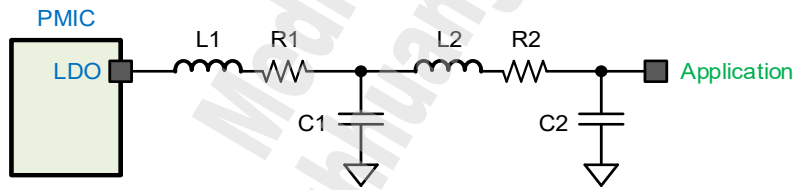
LDO layout constraint for stability								
LDO name	$L1 \leq$	$R1 \leq$	$C1_{min}$	$L2 \leq$	$R2 \leq$	$C2_{min}$	$C2_{max}$	$(C1+C2)_{max}$
VA12_1	20 nH	0.2Ω	1 μ F	20 nH	0.2Ω	0 μ F	10 μ F	11 μ F
VA12_2	20 nH	0.2Ω	1 μ F	20 nH	0.2Ω	0 μ F	10 μ F	11 μ F
VCN13	7 nH	0.2Ω	4.7 μ F	20 nH	0.2Ω	4.7 μ F	14.4 μ F	19.1 μ F
VRF13	7 nH	0.2Ω	4.7 μ F	20 nH	0.2Ω	4.7 μ F	15 μ F	19.7 μ F
VRF12	20 nH	0.2Ω	2.2 μ F	20 nH	0.2Ω	1 μ F	4.4 μ F	6.6 μ F
VUFS12 (Local Sense)	10 nH	0.2Ω	4.7 μ F	20 nH	0.2Ω	4.7 μ F	39.3 μ F	44 μ F
VSRAM_CPUB	7 nH	0.2Ω	2.2 μ F	20 nH	0.2Ω	2.2 μ F	3.61 μ F	5.81 μ F
VSRAM_CPUM	7 nH	0.2Ω	2.2 μ F	20 nH	0.2Ω	2.2 μ F	3.61 μ F	5.81 μ F
VSRAM_CPUL	10 nH	0.1Ω	4.7 μ F	10 nH	0.05Ω	2.2 μ F	4.7 μ F	9.4 μ F



Note: This constraint for LDO stability; application PCB IR should be considered by another check.

LDO: Layout Constraint for LDO Stability (3/3)

LDO layout constraint for stability								
LDO name	$L1 \leq$	$R1 \leq$	$C1_{min}$	$L2 \leq$	$R2 \leq$	$C2_{min}$	$C2_{max}$	$(C1+C2)_{max}$
VRF09	10 nH	0.1Ω	10 uF	20 nH	0.2Ω	9.4 uF	14.1 uF	24.1 uF
VSRAM_MODEM	10 nH	0.1Ω	4.7 uF	10 nH	0.05Ω	2.2 uF	4.7 uF	9.4 uF
VSRAM_DIGRF	7 nH	0.2Ω	2.2 uF	20 nH	0.2Ω	2.2 uF	3.61 uF	5.81 uF
VSRAM_MDFE	7 nH	0.2Ω	2.2 uF	20 nH	0.2Ω	2.2 uF	3.61 uF	5.81 uF
VSRAM_APU	7 nH	0.2Ω	2.2 uF	20 nH	0.2Ω	2.2 uF	4.3 uF	6.5 uF
VDIG18	20 nH	0.2Ω	1 uF					1 uF
VAUX18	20 nH	0.2Ω	1 uF	20 nH	0.2Ω	0 uF	1.2 uF	2.2 uF
VTREF18	20 nH	0.2Ω	1 uF	20 nH	0.2Ω	0 uF	1.2 uF	2.2 uF
VEMC (UFS 3.1)	20 nH	0.2Ω	1 uF	20 nH	0.2Ω	4.7 uF	19 uF	20 uF
VEMC (UFS 4.0)	20 nH	0.2Ω	10 uF	20 nH	0.2Ω	10 uF	14.7 uF	24.7 uF



Note: This constraint for LDO stability; application PCB IR should be considered by another check.

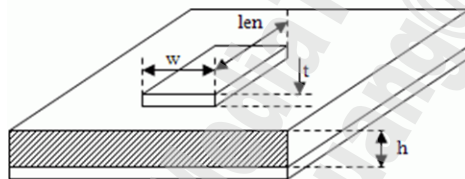
LDO: Rough Guidelines for CAD Extraction

W (mil)	W (um)	H (um)	t @1/3 oz (um)	Len (cm)	Trace inductance (nH)	Trace resistance (mΩ)
4	101.6	50	12	1.27	4.85	198

$$L \text{ (nH)} = 4.85 \text{ nH} \times \frac{101.6 \text{ um}}{w} \times \frac{\text{len}}{1.27 \text{ cm}} \times \frac{h}{50 \text{ um}}$$

$$R \text{ (m}\Omega\text{)} = 198 \text{ m}\Omega \times \frac{101.6 \text{ um}}{w} \times \frac{\text{len}}{1.27 \text{ cm}}$$

where $w = 4\text{mil} \sim 40\text{mil}$



Parameter	Value
Dielectrically constant	4.0
Conductivity	5.8e7 s/m

w = trace width
 len = trace length
 t = trace thickness
 h = dielectric height

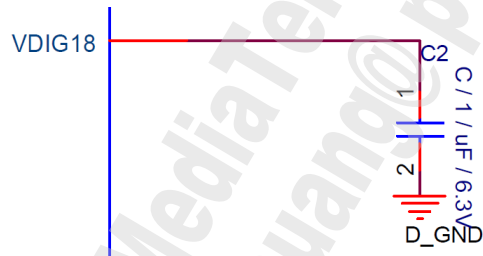
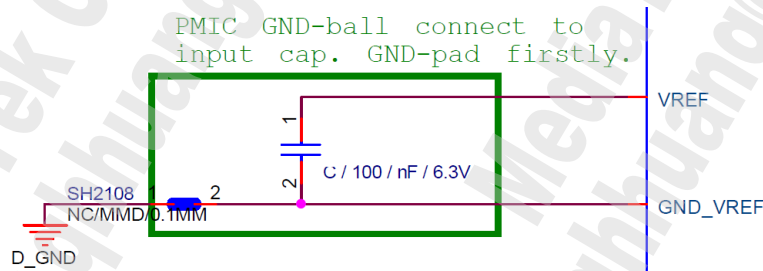
LDO: Layout Constraint for LDO Need Star-connection

- **Some LDO applications need star-connection.**

- VIO18
- VA15
- VA12_1
- VA12_2
- VUFS12
- VCN15
- VRF18
- VRF13
- VRF12

LDO: Layout Constraint for VREF/VDIG18

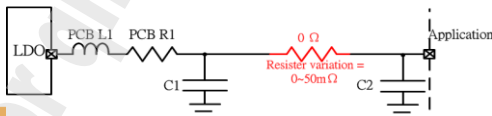
- Place VREF/VDIG18 bypass capacitor as close as possible to the PMIC.
- GND_VREF ball must be connected to the capacitor GND pad firstly, and connect to main GND plane by via.
- VREF capacitor is 100 nF.
- VDIG18 capacitor is 1 uF.
- Schematic example:



LDO: Layout Constraint

- **Follow MES**
 - VA15, VA12_1, VA12_2, VUFS12
 - VSRAM_xxx
- **Trace Under shielding case**
 - VRF18, VCN15, VRFIO18
 - VRF13, VRF12, VCN13
 - VRF09, VRF09_S
- **Local Sense**
 - VUFS12, VUFS12_S
- **Remote sense to application cap.**
 - VSRAM_CPUL
 - VSRAM_MODEM_S, VRF09_S
- **Trace W/L can be adjusted by Imax.**

If trace is series 0Ω resistor, 0Ω resistor is having 0~50mΩ variation. It would be drop voltage.



1. Trace width $\geq 6\text{mil}$

2. Value and placement of capacitor please refer design notice

Ball name	LDO Imax	Trace Length	Trace Width	Rpcb H=50um, 1/3 oz
VAUX18	20mA	800mil	6mil	200mΩ
VTREF	10mA	800mil	6mil	200mΩ
VEMC (if used)	800mA	1200mil	18mil	100mΩ
VIO18	600mA	1000mil	25mil	60mΩ
VA15	600mA	850mil	25mil	50mΩ
VM18	600mA	750mil	25mil	45mΩ
VRF18	600mA	1200mil	18mil	100mΩ
VCN15	300mA	800mil	6mil	200mΩ
VRFIO18	600mA	800mil	6mil	200mΩ
VUFS18	1200mA	1000mil	6mil	200mΩ
VA12_1	600mA	680mil	25mil	40mΩ
VA12_2	600mA	680mil	25mil	40mΩ
VCN13	800mA	1200mil	18mil	100mΩ
VRF13	800mA	1200mil	18mil	100mΩ
VRF12	600mA	1000mil	10mil	150mΩ
VUFS12 #1	1200mA	680mil	25mil	40mΩ
VSRAM_CPUB	600mA	1200mil	18mil	100mΩ
VSRAM_CPUM	600mA	1200mil	18mil	100mΩ
VSRAM_CPUL #1	1200mA	120mil	10mil	18mΩ
VSRAM_MODEM #1	1200mA	75mil	10mil	11mΩ
VSRAM_MDFF	600mA	1200mil	18mil	100mΩ
VSRAM_DIGRF	600mA	850mil	10mil	130mΩ
VSRAM_APU	600mA	1200mil	18mil	100mΩ
VRF09 #1	800mA	120mil	18mil	10mΩ

Note #1: Rpcb from remote sense to application cap.

MT6363 Design Notice

Analog Power PCB Layout Constraint

AP Analog Power List

Voltage	PMIC Pin Name	AP analog pin name	
1.5V	MT6363: VA15	AVDD15_UFS AVDD15_DSI AVDD15_SSUSB_P1 AVDD15_SSUSB_P2 AVDD15_USB AVDD15_CKSQ AVDD15_DRF	AVDD15_APPLL AVDD15_APUPLL AVDD15_MDPLL AVDD15_EMI0 AVDD15_EMI1 AVDD15_EMI2 AVDD15_EMI3 AVDD15_WBG AVDD15_PCIE AVDD15_CSI AVDD15_PROC
0.75V	MT6363: BUCK4 (DVDD_SRAM_CORE)	AVDD08_DRV_DSI	
3.0V	MT6373: VUSB	AVDD33_USB	

AP Analog Power List

Voltage	PMIC Pin Name	AP analog pin name		
1.2V	MT6363: VA12_1	AVDD12_DSI AVDD12_USB AVDD12_CKSQ AVDD12_APPLL AVDD12_APUPLL AVDD12_MDPLL AVDD12_EMI0 AVDD12_EMI1 AVDD12_EMI2 AVDD12_EMI3 AVDD12_TSFDC_BM AVDD12_TSFDC_RM	AVDD12_SSUSB_P1 AVDD12_SSUSB_P2	
1.2V	MT6363: VA12_2	AVDD12_UFS_RX	AVDD12_UFS_TX	
1.2V	MT6363: VUFS12	AVDD12_WBG AVDD12_PCIE	AVDD12_CSI0	AVDD12_CSI1
1.2V	MT6363: VRF12	AVDD12_DRF	AVDD12_DRF (MT6195)	
1.2V	MT6685: VBBCK	AVDD12_CKBUFF_UFS		

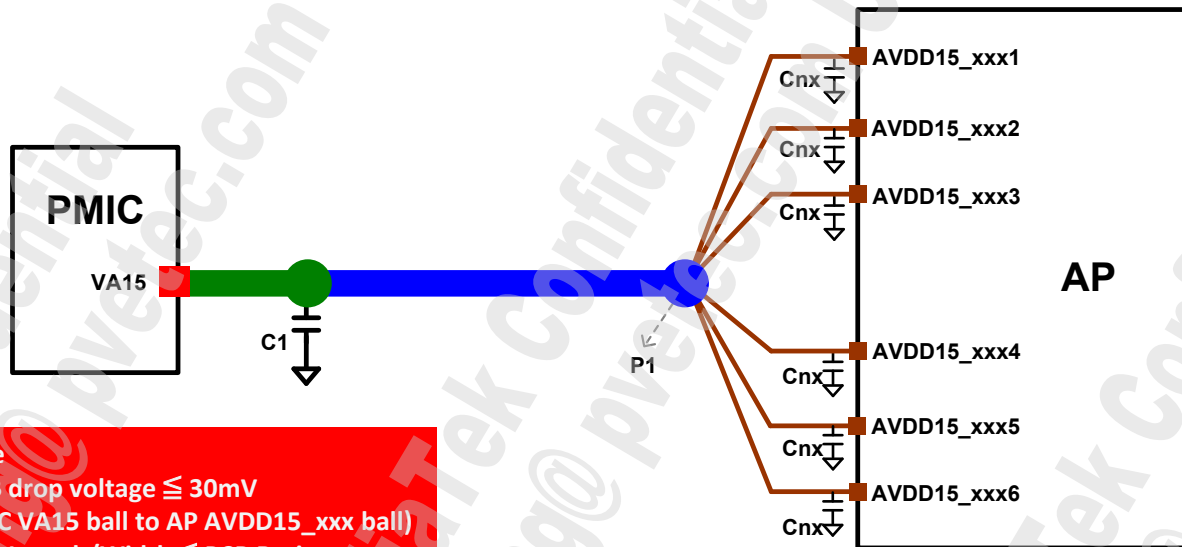
◆ AVDD12_DRF layout constraint

-
- The diagram shows the PCB layout for the VREF12 and VREF13 PMU and FE star connections. The layout includes the following components and connections:
- Top Left:** Labels for VREF12_S, H14, H12, E12, and D12.
 - Top Right:** Labels for VREF12_PMU and AVDD12_DRF_D star connect. Short pad route from PMIC side Cap.
 - Center:** A series of capacitors (C2130, C2129, C2128, C2127, C2123, C2122, C2121, C2120) connected to a common ground plane. Each capacitor is labeled with its value and voltage rating (e.g., C2130 1uF/16.3V).
 - Bottom Left:** Labels for VREF13 and D12.
 - Bottom Right:** A detailed view of the star connection for VREF13_PMU and VREF13_FE. It shows a central node connected to four capacitors (SH2122, SH2121, SH2123, SH2124) which are then connected to the respective PMU and FE pins. The labels for these pins are VREF12_PMU, VREF13_PMU, and VREF13_FE.
 - Bottom Right Text:** VREF13_PMU and VREF13_FE star connect. Short pad route from PMIC side Cap.

◆ IR spec.

- Total path from PMIC LDO pin to AVDD power pin of AP-site
- **Target IR < 2%**
 - For each AVDDxx_xxxx path
 - Simulate all power pin currents at the same time

AP VA15 → AVDD15_xxx Layout Guideline 1



Notice

1. PCB drop voltage $\leq 30\text{mV}$
(PMIC VA15 ball to AP AVDD15_xxx ball)
2. PCB Length/Width $\leq$ PCB Ratio

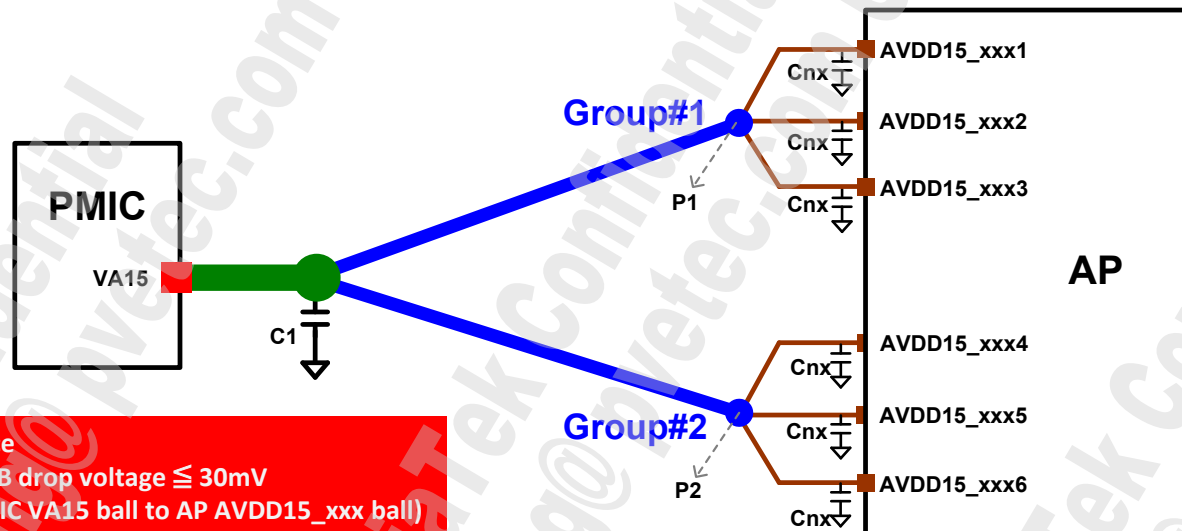
Example			Average current	PCB Width	PCB Length	PCB Ratio Length/Width	PCB Thickness	PCB Resister	PCB drop voltage
Trace	Trace Start	Trace End							
Trace 1	PMIC VA15 ball	C1	0.300A	30mi1	200mi1	7	0.3oz	11.8m	3.5mV
Trace 2	C1	P1	0.300A	18mi1	800mi1	44	0.3oz	78.8m	23.7mV
Trace 3	P1	AP AVDD15_XXX ball	0.060A	8mi1	200mi1	25	0.3oz	44.3m	2.7mV

Pass $\leq 30\text{mV}$

AP VA15 → AVDD15_xxx Layout Guideline 2

Group#1: AVDD15_UFS, AVDD15_DSI, AVDD15_SSUSB_P1, AVDD15_SSUSB_P2, AVDD15_USB, AVDD15_CKSQ, AVDD15_DRF

Group#2: AVDD15_APPLL, AVDD15_APUPLL, AVDD15_MDPLL, AVDD15_EMI0, AVDD15_EMI1, AVDD15_EMI2, AVDD15_EMI3, AVDD15_WBG, AVDD15_PCIE, AVDD15_CSI, AVDD15_PROC



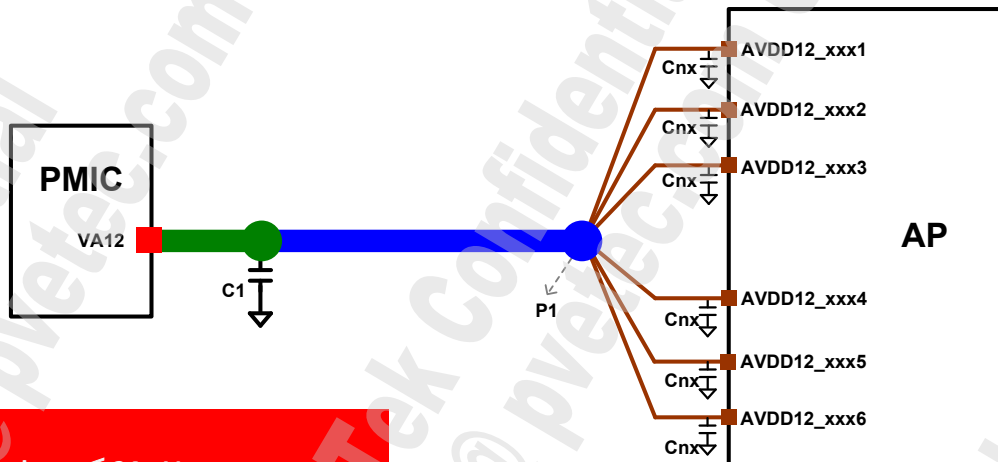
Notice

1. PCB drop voltage $\leq 30\text{mV}$
(PMIC VA15 ball to AP AVDD15_xxx ball)
2. PCB Length/Width $\leq$ PCB Ratio

Example			Average current	PCB Width	PCB Length	PCB Ratio Length/Width	PCB Thickness	PCB Resistor	PCB drop voltage
Trace	Trace Start	Trace End							
Trace 1	PMIC VA15 ball	C1	0.300A	24mi1	200mi1	8	0.3oz	14.8m	4.4mV
Trace 2	C1	P1/P2	0.150A	12mi1	1000mi1	83	0.3oz	147.8m	22.2mV
Trace 3	P1/P2	AP AVDD15_XXX ball	0.060A	8mi1	250mi1	31	0.3oz	55.4m	3.3mV

Pass $\leq 30\text{mV}$ Right © MediaTek Inc. All rights reserved.

AP VA12_1 → AVDD12_xxx Layout Guideline 1



Notice

1. PCB drop voltage $\leq 24\text{mV}$
(PMIC VA12_1 ball to AP AVDD12_xxx ball)
2. PCB Length/Width $\leq$ PCB Ratio

	Example		Average current	PCB Width	PCB Length	PCB Ratio Length/Width	PCB Thickness	PCB Resister	PCB drop voltage
Trace 1	PMIC VA12_1 ball	C1	0.500A	36mi l	200mi l	6	0.3oz	9.9m	4.9mV
Trace 2	C1	P1	0.500A	32mi l	550mi l	17	0.3oz	30.5m	15.2mV
Trace 3	P1/P2	AP AVDD12_XXX ball	0.130A	12mi l	200mi l	17	0.3oz	29.6m	3.8mV

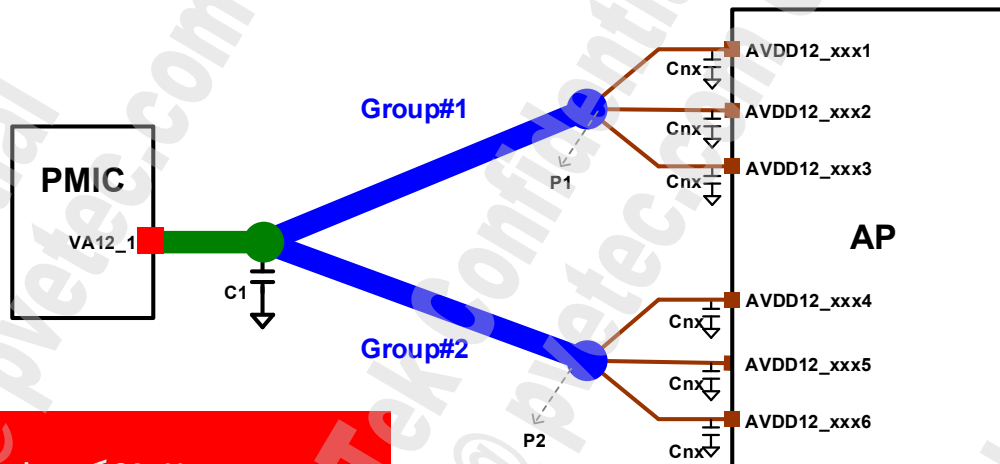


Pass $\leq 24\text{mV}$

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AP VA12_1 → AVDD12_xxx Layout Guideline 2

Group#1: AVDD12_DSI, AVDD12_USB, AVDD12_CKQ, AVDD12_APPLL, AVDD12_APUPLL, AVDD12_MDPLL, AVDD12_EMI0, AVDD12_EMI1, AVDD12_EMI2, AVDD12_EMI3, AVDD12_TSFDC_BM, AVDD12_TSFDC_RM
Group#2: AVDD12_SSUSB_P1, AVDD12_SSUSB_P2



Notice

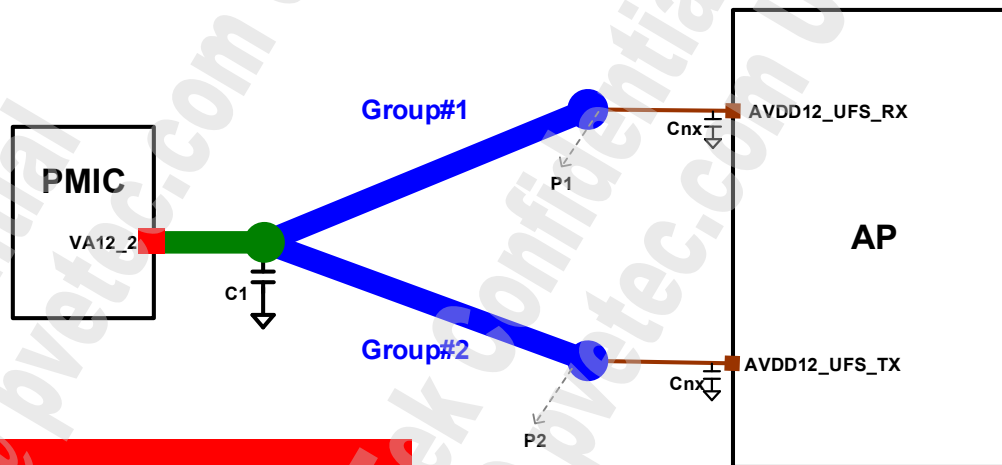
1. PCB drop voltage $\leq 24\text{mV}$
(PMIC VA12_1 ball to AP AVDD12_xxx ball)
2. PCB Length/Width $\leq$ PCB Ratio

	Example		Average current	PCB Width	PCB Length	PCB Ratio Length/Width	PCB Thickness	PCB Resister	PCB drop voltage
Trace	Trace Start	Trace End							
Trace 1	PMIC VA12_1 ball	C1	0.500A	36mi l	150mi l	4	0.3oz	7.4m	3.7mV
Trace 2	C1	P1/ P2	0.260A	24mi l	750mi l	31	0.3oz	55.4m	14.4mV
Trace 3	P1/ P2	AP AVDD12_XXX ball	0.130A	8mi l	200mi l	25	0.3oz	44.3m	5.8mV

AP VA12_2 → AVDD12_XXX Layout Guideline

Group#1: AVDD12_UFS_RX

Group#2: AVDD12_UFS_TX



Notice

1. PCB drop voltage $\leq 24\text{mV}$
(PMIC VA12_2 ball to AP AVDD12_XXX ball)
2. PCB Length/Width $\leq$ PCB Ratio

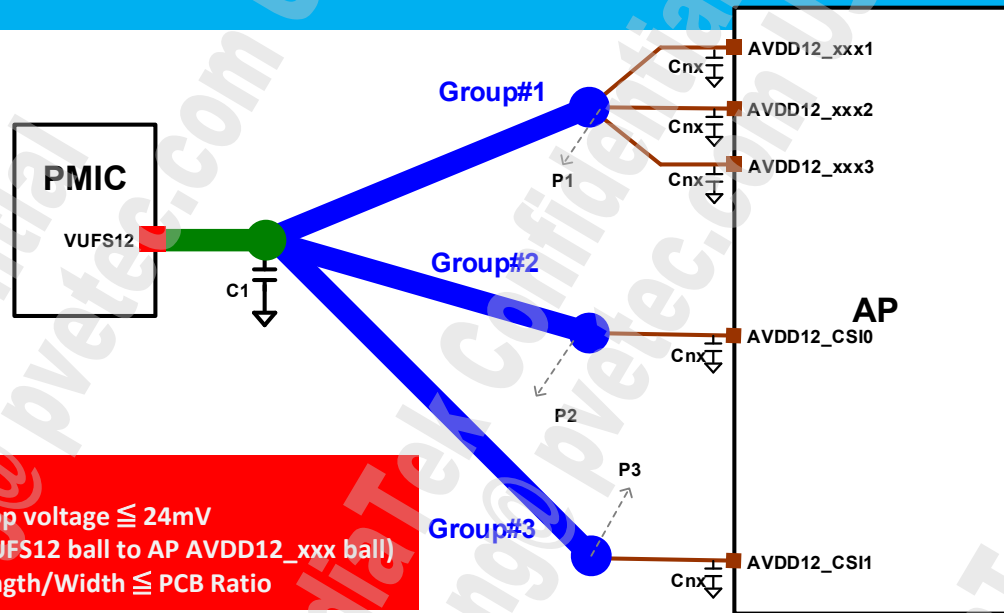
Example			Average current	PCB Width	PCB Length	PCB Ratio Length/Width	PCB Thickness	PCB Resister	PCB drop voltage
Trace	Trace Start	Trace End							
Trace 1	PMIC VA12_2 ball	C1	0.320A	36mil	200mil	6	0.3oz	9.9m	3.2mV
Trace 2	C1	P1/P2	0.180A	20mil	800mil	40	0.3oz	71.0m	12.8mV
Trace 3	P1/P2	AP AVDD12_UFS_XX ball	0.180A	8mil	200mil	25	0.3oz	44.3m	8.0mV

AP VUFS12 → AVDD12_XXX Layout Guideline

Group#1: AVDD12_WBG, AVDD12_PCIE

Group#2: AVDD12_CSI0

Group#3: AVDD12_CSI1



Notice

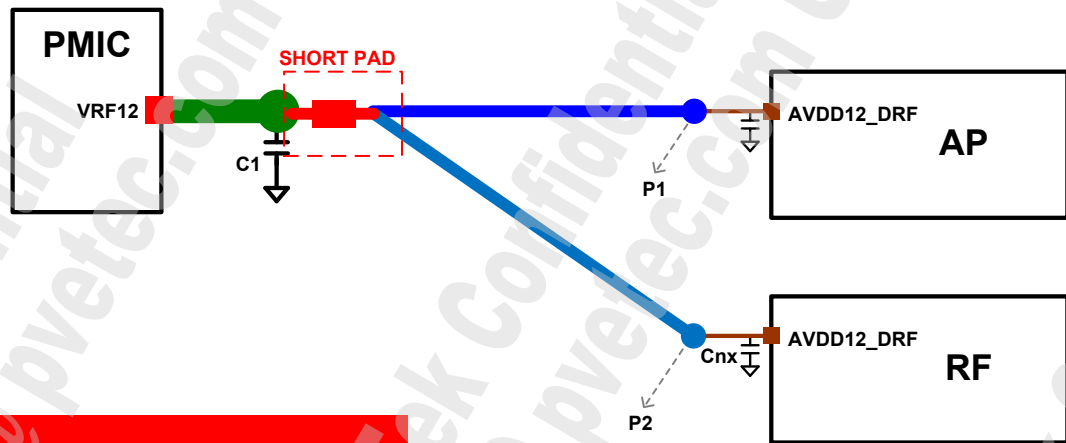
1. PCB drop voltage $\leq 24\text{mV}$
(PMIC VUFS12 ball to AP AVDD12_XXX ball)
2. PCB Length/Width $\leq$ PCB Ratio

Example			Average current	PCB Width	PCB Length	PCB Ratio Length/Width	PCB Thickness	PCB Resister	PCB drop voltage
Trace	Trace Start	Trace End							
Trace 1	PMIC VUFS12 ball	C1	0.660A	36mi l	150mi l	4	0.3oz	7.4m	4.9mV
Trace 2	C1	P1/ P2/ P3	0.250A	32mi l	850mi l	27	0.3oz	47.1m	11.8mV
Trace 3	P1/ P2/ P3	AP AVDD12_XXX ball	0.250A	12mi l	200mi l	17	0.3oz	29.6m	7.4mV



Pass $\leq 24\text{mV}$ MediaTek Inc. All rights reserved.

AP VRF12 → AVDD12_xxx Layout Guideline



Notice

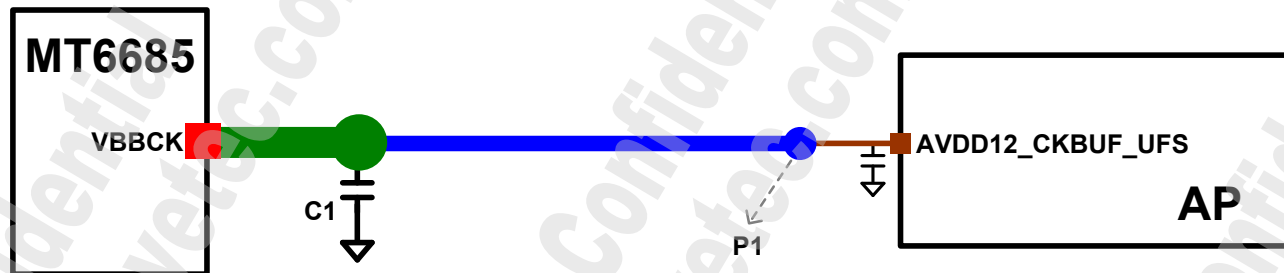
1. PCB drop voltage $\leq 24\text{mV}$
(PMIC VRF12 ball to AP AVDD12_DRF ball)
2. PCB Length/Width $\leq$ PCB Ratio

Example			Average current	PCB Width	PCB Length	PCB Ratio Length/Width	PCB Thickness	PCB Resistor	PCB drop voltage
Trace	Trace Start	Trace End							
Trace 1	PMIC VRF12 ball	C1	0.280A	24mil	100mil	4	0.3oz	7.4m	2.1mV
Trace 2	C1	P1/P2	0.140A	16mil	1000mil	63	0.3oz	110.9m	15.5mV
Trace 3	P1/P2	AP/RF AVDD12_DRF	0.140A	8mil	200mil	25	0.3oz	44.3m	6.2mV



Pass $\leq 24\text{mV}$

AP AVDD12_CKBUF_UFS Layout Guideline

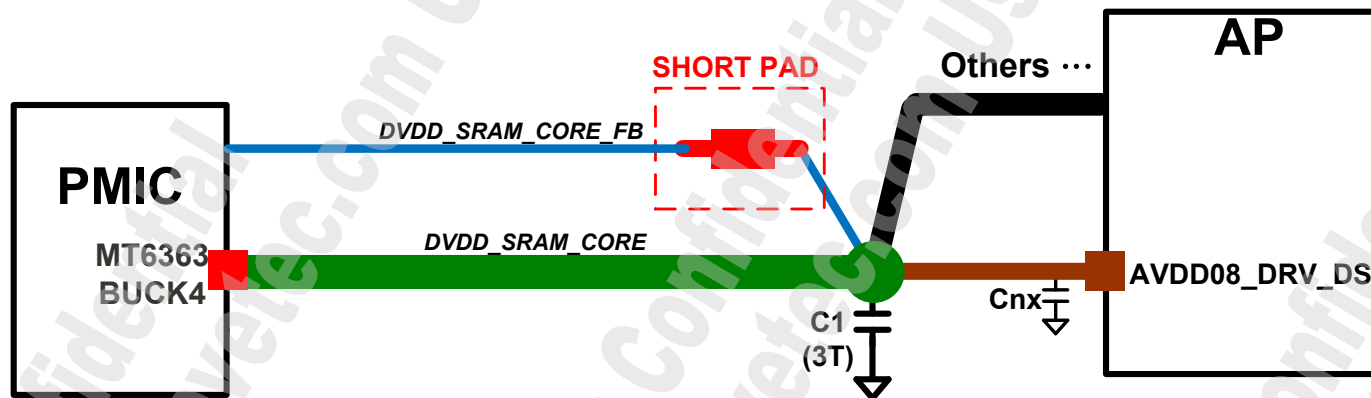


Notice

1. PCB drop voltage $\leq 24\text{mV}$
(VBBCK ball to AP AVDD12_CKBUF_UFS ball)
2. PCB Length/Width $\leq$ PCB Ratio

Example			Average current	PCB Width	PCB Length	PCB Ratio Length/Width	PCB Thickness	PCB Resister	PCB drop voltage
Trace	Trace Start	Trace End							
Trace 1	PMIC VBBCK	C1	0.020A	8mil	200mil	25	0.3oz	44.3m	0.9mV
Trace 2	C1	P1	0.020A	8mil	1200mil	150	0.3oz	266.1m	5.3mV
Trace 3	P1	AP AVDD12_CKBUF_UFS	0.020A	8mil	200mil	25	0.3oz	44.3m	0.9mV

AP AVDD08_DRV_DS Layout Guideline



Notice

1. PCB drop voltage $\leq 15\text{mV}$
(SRAM_CORE ball to AVDD08_DRV_DS ball)
2. PCB Length/Width $\leq$ PCB Ratio

Trace	Example		Average current	PCB W dth	PCB Length	PCB Ratio Length/W dth	PCB Thickness	PCB Resister	PCB drop voltage
	Trace Start	Trace End							
Trace 1	C1	AP AVDD08_DRV_DS ball	0.050A	8mil	1200mil	150	0.3oz	266.1m	13.3mV

Pass $\leq 15\text{mV}$

AP AVDD33_USB Layout Guideline



Notice

1. PCB drop voltage $\leq 30\text{mV}$
(PMIC VUSB ball to AP AVDD33_USB ball)
2. PCB Length/Width $\leq$ PCB Ratio

Example			Average current	PCB Width	PCB Length	PCB Ratio Length/Width	PCB Thickness	PCB Resistor	PCB drop voltage
Trace	Trace Start	Trace End							
Trace 1	PMIC VUSB ball	C1	0.040A	12mil	1000mil	83	0.3oz	147.8m	5.9mV
Trace 2	C1	AP AVDD33_USB ball	0.040A	8mil	1000mil	125	0.3oz	221.7m	8.9mV

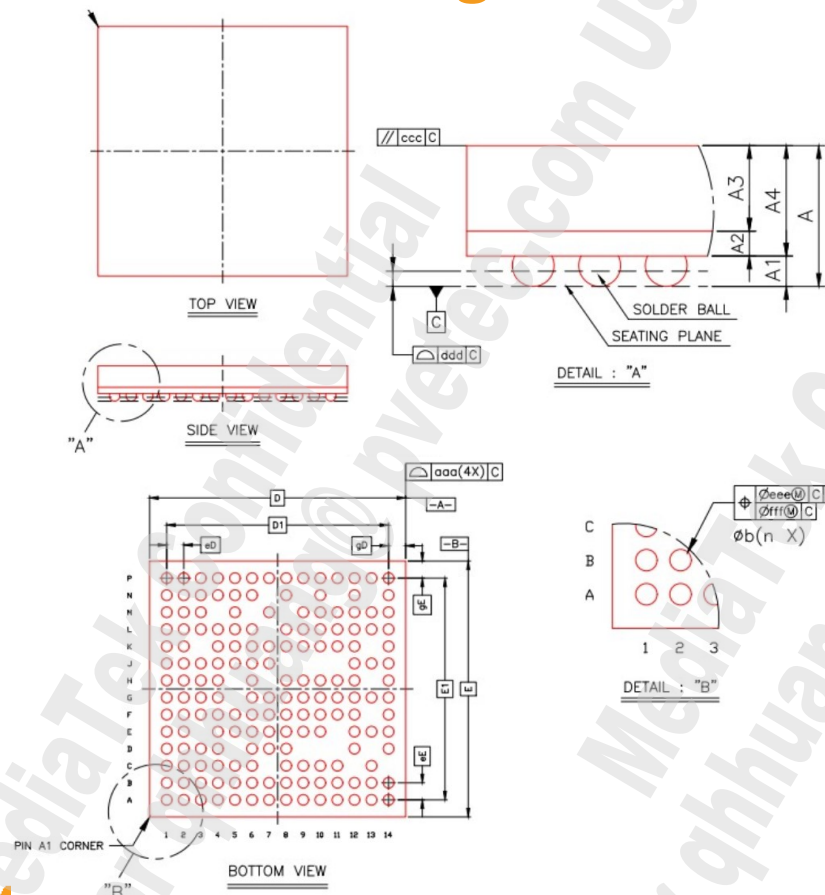
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MT6363 Design Notice

Package

MT6363: Package Outline



Item		Symbol	Common Dimensions		
			MIN.	NOM.	MAX.
Package Type			MFC WFBGA		
Body Size	X	D	5.20	5.25	5.30
	Y	E	5.20	5.25	5.30
Ball Pitch	X	eD	0.35		
	Y	eE	0.35		
Mold Thickness		A3	0.45 Ref.		
Substrate Thickness		A2	0.13 Ref.		
Substrate+Mold Thickness		A4	0.53	0.58	0.63
Total Thickness		A	—	—	0.80
Ball Diameter			0.22		
Ball Stand Off		A1	0.12	0.16	0.20
Ball Width		b	0.17	0.22	0.27
Package Edge Tolerance		aaa	0.05		
Mold Flatness		ccc	0.10		
Coplanarity		ddd	0.08		
Ball Offset (Package)		eee	0.15		
Ball Offset (Ball)		fff	0.05		
Ball Count		n	167		
Edge Ball Center to Center	X	D1	4.55		
	Y	E1	4.55		
Edge Ball Center to Package Edge	X	gD	0.35		
	Y	gE	0.35		

MT6363: Ball-map

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
A	VSYS_BUCK7	VSYS_BUCK7	VSYS_BUCK3	BUCK3	GND_BUCK3	GND_BUCK2	BUCK2	VSYS_BUCK2	VSYS_BUCK1	BUCK1	GND_BUCK1	GND_VS2	VS2	VSYS_VS2	A
B	BUCK7	BUCK7	VSYS_BUCK3	BUCK3	GND_BUCK3	GND_BUCK2	BUCK2	VSYS_BUCK2	VSYS_BUCK1	BUCK1	GND_BUCK1	GND_VS2	VS2	VSYS_VS2	B
C	GND_BUCK7	GND_BUCK7	GND_BUCK7_FB	VSYS_SMPS	GND_BUCK3_FB	BUCK3_FB		BUCK2_FB	GND_BUCK2_FB	GND_BUCK1_FB	BUCK1_FB		VS2_FB		C
D	GND_BUCK6	GND_BUCK6	BUCK7_FB	GND_SMPS		GND	GND	GND				VRF13		VUFS12	D
E	BUCK6	BUCK6		SRCLKEN_IN0	DVDD18_IO	GND	GND	GND	GPIO7	EINT		VRF12	VS2_LDO1	VUFS12_S	E
F	VSYS_BUCK6	VSYS_BUCK6	BUCK6_FB	GPIO8	SRCLKEN_IN1	GND	GND	GND	GPIO6	SCP_VREQ_VAO	RTC_32KIN	VSRAM_CPU_M		VSRAM_CPUB	F
G	VSYS_BUCK5	VSYS_BUCK5	GND_BUCK6_FB	GPIO9		GND	GND	GND	GPIO4	GPIO2	WDTRSTB_IN	VSRAM_CPUL	VSRAM_CPUL_S	VCN13	G
H	BUCK5	BUCK5	GND_BUCK5_FB	FSOURCE		SPMI_M_SDA		SPMI_M_SCL	GPIO5	GPIO3	GPIO1	VA12_2		VA12_1	H
J	GND_BUCK5	GND_BUCK5	BUCK5_FB	EXT_PMIC_EN2	EXT_PMIC_PG	RTC_INT	VDIG18					VAUX18	VS2_LDO2	VEMC	J
K	GND_BUCK4	GND_BUCK4		EXT_PMIC_EN1	SYSRSTB	VSYSSENS	CHRDETB	GND_VREF	GND	GND	GND	VTREF18		VSYS_LDO1	K
L	BUCK4	BUCK4	BUCK4_FB	PMU_TESTMODE	PWRKEY	VBBSNS		VREF	GND	GND	AVSS18_AUXADC	VIO18	VS1_LDO2	VUFS18	L
M	VSYS_BUCK4	VSYS_BUCK4	GND_BUCK4_FB		VS1_FB		VS3_FB		VSRAM_APU	VRF09	VRF09_S	VCN15	VRFIO18	VIO075	M
N	VSYS_VS1	VS1	GND_VS1	GND_VS3	VS3	VSYS_VS3		VS3_LDO1		VS3_LDO2		VS1_LDO1		GND	N
P	VSYS_VS1	VS1	GND_VS1	GND_VS3	VS3	VSYS_VS3	VSRAM_MDFE	VSRAM_DIGRF	VSRAM_MODEM_S	VSRAM_MODEM	VM18	VRF18	VA15	NC	P
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	

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